

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277945

Kind Code

A1

Publication Date

September 04, 2025

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OPTICAL MODULE

Abstract

An optical module includes a light emitting assembly. The light emitting assembly includes a plurality of lasers, a plurality of wavelength division multiplexers and a lens group. The plurality of lasers emit a plurality of optical signals. The plurality of wavelength division multiplexers multiplex the plurality of optical signals into a plurality of composite optical signals. The lens group includes a first lens, a second lens, and a third lens. The second lens is configured to transmit a first part of the plurality of composite optical signals exited from the first lens, reflect a second part of the plurality of composite optical signals exited from the third lens to the first lens, and transmit the second part of the plurality of composite optical signals reflected by the first lens, so as to multiplex the plurality of composite optical signals into the merge composite optical signal.

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Family ID: 80775724

Appl. No.: 19/211924

Filed: May 19, 2025

Foreign Application Priority Data

CN	202010988200.8	Sep. 18, 2020
CN	202010988545.3	Sep. 18, 2020
CN	202010988607.0	Sep. 18, 2020
CN	202010990182.7	Sep. 18, 2020

Related U.S. Application Data

Publication Classification

Int. Cl.: **G02B6/42** (20060101); **G02B6/27** (20060101); **G02B6/293** (20060101); **G02B6/43** (20060101)

U.S. Cl.:

CPC **G02B6/4215** (20130101); **G02B6/2773** (20130101); **G02B6/2938** (20130101); **G02B6/423** (20130101); **G02B6/426** (20130101); **G02B6/43** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. patent application Ser. No. 18/122,560 (published as US2023/0258883 A1), which is a continuation-in-part Application of International Patent Application No. PCT/CN2020/135775, filed on Dec. 11, 2020, which claims priorities to Chinese Patent Application No. 202010988200.8, filed on Sep. 18, 2020; Chinese Patent Application No. 202010990182.7, filed on Sep. 18, 2020; Chinese Patent Application No. 202010988545.3, filed on Sep. 18, 2020; and Chinese Patent Application No. 202010988607.0, filed on Sep. 18, 2020, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of optical communication technologies, and in particular, to an optical module.

BACKGROUND

[0003] With the development of new services and application scenarios such as cloud computing, mobile Internet and video, development and progress of optical communication technology has become increasingly important. In optical communication technology, an optical module is a tool for achieving interconversion between an optical signal and an electrical signal and is one of key components in an optical communication device. In addition, with the development of optical communication technology, it is required that a transmission rate of the optical module is continuously increasing.

SUMMARY

[0004] In an aspect, an optical module is provided. The optical module includes a circuit board, a light emitting device, a fiber optic adapter and a light receiving device. The light emitting device is electrically connected with the circuit board. The light emitting device includes a light emitting housing, a light emitting cover plate, a plurality of lasers, a plurality of wavelength division multiplexers and a lens group. The light emitting cover plate is covered on the light emitting housing and defines a light emitting cavity with the light emitting housing. The plurality of lasers are disposed in the light emitting cavity and are configured to emit a plurality of optical signals with different wavelengths. The plurality of wavelength division multiplexers are disposed in the light emitting cavity and correspond to the plurality of lasers. The plurality of wavelength division multiplexers are configured to multiplex the plurality of optical signals into a plurality of composite optical signals. The plurality of wavelength division multiplexers include a first part of the plurality of wavelength division multiplexers and a second part of the plurality of wavelength

division multiplexers. The lens group is disposed in the light emitting cavity and is configured to multiplex the plurality of composite optical signals into a merge composite optical signal. The lens group includes a first lens, a second lens, and a third lens. The first lens is disposed on light-exit paths of the first part of the plurality of wavelength division multiplexers. The second lens is disposed on a light-exit path of the first lens. The third lens is disposed on light-exit paths of the second part of the plurality of wavelength division multiplexers. The fiber optic adapter is connected with the light emitting housing and is configured to receive and transmit the merge composite optical signal. The light receiving device is electrically connected with the circuit board and is configured to receive optical signals from an outside of the optical module. The light receiving device and the light emitting device are arranged in a stack. The plurality of composite optical signals include a first part of the plurality of composite optical signals and a second part of the plurality of composite optical signals. The second lens is configured to transmit the first part of the plurality of composite optical signals exited from the first lens, reflect the second part of the plurality of composite optical signals exited from the third lens to the first lens, and transmit the second part of the plurality of composite optical signals reflected by the first lens to adjust angles incident on the second lens, so as to multiplex the plurality of composite optical signals into the merge composite optical signal.

[0005] In another aspect, another optical module is provided. The optical module includes a circuit board, a light emitting device, a fiber optic adapter and a light receiving device. The light emitting device is electrically connected with the circuit board. The light emitting device includes a light emitting housing, a light emitting cover plate, a plurality of lasers and a lens group. The light emitting cover plate is covered on the light emitting housing and defines a light emitting cavity with the light emitting housing. The plurality of lasers are disposed in the light emitting cavity, and are configured to emit a plurality of optical signals with different wavelengths. The lens group is disposed in the light emitting cavity, and is configured to multiplex the plurality of optical signals into a merge composite optical signal. The lens group includes a plurality of lenses, a first polarizer, a second polarizer, a polarizing beam splitter and a first lozenge lens. The plurality of lenses are respectively located on light-exit paths of the plurality of lasers and are configured to multiplex the plurality of optical signals into a plurality of composite optical signals. The plurality of lenses include a first part of the plurality of lenses and a second part of the plurality of lenses. The plurality of composite optical signals include a first part of the plurality of composite optical signals exited from the first part of the plurality of lenses and a second part of the plurality of composite optical signals exited from the second part of lenses. The first polarizer is located on light-exit paths of the first part of the plurality of lenses and is configured to convert the first part of the plurality of composite optical signals into polarized lights with a polarization direction of a first direction. The second polarizer is located on light-exit paths of the second part of the plurality of lenses, and is configured to convert the second part of the plurality of composite optical signals into polarized lights with a polarization direction of a second direction. The first direction is perpendicular to the second direction. The polarizing beam splitter includes a splitter slope. The splitter slope is located on a light-exit path of the first polarizer. The first lozenge lens is configured to reflect the second part of the plurality of composite optical signals exiting the second polarizer to the splitter slope. The fiber optic adapter is connected with the light emitting housing and is configured to receive and transmit the merge composite optical signal. The light receiving device is electrically connected with the circuit board and is configured to receive optical signals from an outside of the optical module. The light receiving device and the light emitting device are arranged in a stack. The polarizing beam splitter is configured to transmit the polarized lights with the polarization direction of the first direction and reflect the polarized lights with the polarization direction of the second direction, so as to multiplex the plurality of composite optical signals into the merge composite optical signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] In order to describe the technical solutions of the embodiments of the present invention more clearly, accompanying drawings to be used in the description of some embodiments will be introduced briefly below. However, the accompanying drawings to be described below are merely accompanying drawings of some embodiments of the present disclosure, and a person having ordinary skill in the art may obtain other drawings according to these drawings. In addition, the accompanying drawings to be described below may be regarded as schematic diagrams and are not limitations on an actual size of a product, an actual process of a method and an actual timing of a signal to which the embodiments of the present disclosure relate.

[0007] FIG. 1 is a connection diagram of an optical communication system, in accordance with some embodiments;

[0008] FIG. 2 is a structural diagram of an optical network terminal, in accordance with some embodiments;

[0009] FIG. 3 is a structural diagram of an optical module, in accordance with some embodiments;

[0010] FIG. 4 is an exploded view of an optical module, in accordance with some embodiments;

[0011] FIG. 5 is a local sectional view of an optical module, in accordance with some embodiments;

[0012] FIG. 6 is an exploded view of a light emitting device and a light receiving device in an optical module, in accordance with some embodiments;

[0013] FIG. 7 is an exploded view of a light emitting device in an optical module, in accordance with some embodiments;

[0014] FIG. 8 is a diagram showing a partial structure of a light emitting device in an optical module, in accordance with some embodiments;

[0015] FIG. 9A is a top view showing a portion of structure of a light emitting device in an optical module, in accordance with some embodiments;

[0016] FIG. 9B is an optical path diagram of a light emitting device in an optical module, in accordance with some embodiments;

[0017] FIG. 9C is an optical path diagram of a lens group in an optical module, in accordance with some embodiments;

[0018] FIG. 10A is an optical path diagram of another light emitting device in an optical module, in accordance with some embodiments;

[0019] FIG. 10B is an optical path diagram of another lens group in an optical module, in accordance with some embodiments;

[0020] FIG. 11A is an optical path diagram of yet another light emitting device in an optical module, in accordance with some embodiments;

[0021] FIG. 11B is an optical path diagram of yet another lens group in an optical module, in accordance with some embodiments;

[0022] FIG. 12 is a structural diagram of a lens group, an wavelength division multiplexer and a supporting platform in an optical module, in accordance with some embodiments;

[0023] FIG. 13 is a structural diagram of a supporting platform in an optical module, in accordance with some embodiments;

[0024] FIG. 14 is an enlarged view of cycle A in FIG. 13;

[0025] FIG. 15 is a perspective view of a laser, a heat dissipation component and a ceramic adapter block in an optical module, in accordance with some embodiments;

[0026] FIG. 16 is an exploded view of a laser and a heat dissipation component in an optical module, in accordance with some embodiments;

[0027] FIG. 17 is a structural diagram of an optical detector and a laser driver in an optical module,

in accordance with some embodiments;

[0028] FIG. **18** is an exploded view of an optical detector in an optical module, in accordance with some embodiments;

[0029] FIG. **19** is an exploded view of a light emitting housing and a ceramic adapter block in an optical module, in accordance with some embodiments;

[0030] FIG. **20** is a perspective view of a ceramic adapter block in an optical module, in accordance with some embodiments;

[0031] FIG. **21** is a structural diagram of a light emitting housing, a light emitting cover plate and a connecting block in an optical module, in accordance with some embodiments; and

[0032] FIG. **22** is a partial sectional view of a light emitting device in an optical module, in accordance with some embodiments.

DETAILED DESCRIPTION

[0033] Some embodiments of the present disclosure will be described clearly and completely with reference to the accompanying drawings; however, the described embodiments are merely some but not all embodiments of the present disclosure. All other embodiments obtained on a basis of the embodiments of the present disclosure by a person of ordinary skill in the art shall be included in the protection scope of the present disclosure.

[0034] Unless the context requires otherwise, throughout the specification and the claims, the term “comprise” and other forms thereof such as the third-person singular form “comprises” and the present participle form “comprising” are construed as being open and inclusive, meaning “including, but not limited to.” In the description, the terms such as “one embodiment,” “some embodiments,” “exemplary embodiments,” “example,” “specific example,” or “some examples” are intended to indicate that specific features, structures, materials, or characteristics related to the embodiment(s) or example(s) are included in at least one embodiment or example of the present disclosure. Schematic representations of the above terms do not necessarily refer to the same embodiment(s) or example(s). In addition, specific features, structures, materials, or characteristics may be included in any one or more embodiments or examples in any suitable manner.

[0035] Hereinafter, the terms “first” and “second” are only used for descriptive purposes and are not to be construed as indicating or implying relative importance or implicitly indicating the number of indicated technical features. Therefore, the features defined with “first” and “second” may explicitly or implicitly include one or more of these features. In the description of the embodiments of the present disclosure, the terms “a plurality of” and “the plurality of” each mean two or more unless otherwise specified.

[0036] In the description of some embodiments, the term “connected” and derivatives thereof may be used. The term “connected” should be understood in a broad sense. For example, the term “connected” may represent a fixed connection, a detachable connection, or a one-piece connection, or may represent a direct connection, or may represent an indirect connection through an intermediate medium.

[0037] The phrase “at least one of A, B, and C” has the same meaning as the phrase “at least one of A, B, or C,” and they both include the following combinations of A, B, and C: only A, only B, only C, a combination of A and B, a combination of A and C, a combination of B and C, and a combination of A, B, and C.

[0038] The phrase “A and/or B” includes the following three combinations: only A, only B, and a combination of A and B.

[0039] The use of “applicable to” or “configured to” herein indicates an open and inclusive expression, which does not exclude devices that are applicable to or configured to perform additional tasks or steps.

[0040] The term “about,” “substantially,” and “approximately” as used herein includes a stated value and an average value within an acceptable range of deviation of a particular value determined by a person of ordinary skill in the art, considering measurement in question and errors associated

by measurement of a particular quantity (i.e., limitations of a measurement system).

[0041] The terms “parallel,” “perpendicular,” and “equal” as used herein include the stated conditions and the conditions similar to the stated conditions, and the range of the similar conditions is within the acceptable deviation range, where the acceptable deviation range is determined by a person of ordinary skill in the art in consideration of the measurement in question and the error associated with the measurement of a specific quantity (i.e., the limitation of the measurement system). For example, the term “parallel” includes absolute parallelism and approximate parallelism, and an acceptable range of deviation of the approximate parallelism may be, for example, a deviation within 5°; the term “perpendicular” includes absolute perpendicularity and approximate perpendicularity, and an acceptable range of deviation of the approximate perpendicularity may also be, for example, a deviation within 5°. The term “equal” includes absolute equality and approximate equality, and an acceptable range of deviation of the approximate equality may be, for example, a difference between two equals of less than or equal to 5% of either of the two equals.

[0042] In optical communication technology, light is used to carry information to be transmitted, and an optical signal carrying the information is transmitted to an information processing device such as a computer through an information transmission device such as an optical fiber or an optical waveguide, so as to achieve transmission of the information. Since an optical signal has a characteristic of passive transmission when being transmitted through the optical fiber or the optical waveguide, low-cost and low-loss information transmission may be achieved. In addition, a signal transmitted by the information transmission device such as the optical fiber or the optical waveguide is an optical signal, while a signal that can be recognized and processed by the information processing device such as the computer is an electrical signal. Therefore, in order to establish information connection between the information transmission device such as the optical fiber or the optical waveguide and the information processing device such as the computer, it is necessary to realize interconversion between the electrical signal and the optical signal.

[0043] An optical module implements the interconversion between the optical signal and the electrical signal in the field of optical fiber communication technology. The optical module includes an optical port and an electrical port. The optical module achieves optical communication with the information transmission device such as the optical fiber or the optical waveguide through the optical port and achieves electrical connection with an optical network terminal (e.g., an optical modem) through the electrical port. The electrical connection is mainly used for achieving power supply, transmission of an Inter-integrated circuit (I2C) signal, transmission of data signal and grounding. The optical network terminal transmits an electrical signal to the information processing device such as the computer through a network cable or wireless fidelity (Wi-Fi).

[0044] FIG. 1 is a connection diagram of an optical communication system, in accordance with some embodiments. As shown in FIG. 1, the optical communication system mainly includes a remote server **1000**, a local information processing device **2000**, an optical network terminal **100**, an optical module **200**, an optical fiber **101**, and a network cable **103**.

[0045] An end of the optical fiber **101** is connected to the remote server **1000**, and the other end thereof is connected to the optical network terminal **100** through the optical module **200**. The optical fiber itself can support long-distance signal transmission. For example, several-kilometer (6-kilometer to 8-kilometer) signal transmission. On this basis, ultra-long-distance transmission may be achieved theoretically if a repeater is used. Therefore, in a typical optical communication system, a distance between the remote server **1000** and the optical network terminal **100** may typically reach several kilometers, tens of kilometers, or hundreds of kilometers.

[0046] An end of the network cable **103** is connected to the local information processing device **2000**, and the other end thereof is connected to the optical network terminal **100**. The local information processing device **2000** may include any or more of the following devices: a router, a switch, a computer, a mobile phone, a tablet computer, or a television.

[0047] A physical distance between the remote server **1000** and the optical network terminal **100** is greater than a physical distance between the local information processing device **2000** and the optical network terminal **100**. Connection between the local information processing device **2000** and the remote server **1000** is accomplished through the optical fiber **101** and the network cable **103**, and connection between the optical fiber **101** and the network cable **103** is accomplished through the optical network terminal **100**.

[0048] The optical network terminal **100** includes a housing in a substantially cuboid shape, and an optical module interface **102** and a network cable interface **104** that are disposed on the housing. The optical module interface **102** is configured to connect to the optical module **200**, so that bidirectional electrical signal connection is established between the optical network terminal **100** and the optical module **200**. The network cable interface **104** is configured to connect to the network cable **103**, so that bidirectional electrical signal connection is established between the optical network terminal **100** and the network cable **103**. Connection between the optical module **200** and the network cable **103** is established through the optical network terminal **100**. For example, the optical network terminal **100** transmits electrical signals from the optical module **200** to the network cable **103** and transmits electrical signals from the network cable **103** to the optical module **200**. Therefore, the optical network terminal **100**, as a master monitor of the optical module **200**, can monitor the operation of the optical module **200**. In addition to the optical network terminal **100**, the master monitor of the optical module **200** may further include an optical line terminal (OLT).

[0049] The optical module **200** includes an optical port and an electrical port. The optical port is configured to be connected to the optical fiber **101**, so that bidirectional optical signal connection is established between the optical module **200** and the optical fiber **101**. The electrical port is configured to connect to the optical network terminal **100**, so that bidirectional electrical signal connection is established between the optical module **200** and the optical network terminal **100**. The interconversion between the optical signal and the electrical signal is achieved by the optical module **200**, so that connection between the optical fiber **101** and the optical network terminal **100** is established. For example, optical signals from the optical fiber **101** are converted into electrical signals by the optical module **200**, and then the electrical signals are input into the optical network terminal **100**. Electrical signals from the optical network terminal **100** are converted into optical signals by the optical module **200**, and then the optical signals are input into the optical fiber **101**. Since the optical module **200** is a tool for achieving the interconversion between the optical signal and the electrical signal and has no function of processing data, information does not change in the above photoelectric conversion process.

[0050] A bidirectional signal transmission channel between the remote server **1000** and the local information processing device **2000** is established through the optical fiber **101**, the optical module **200**, the optical network terminal **100**, and the network cable **103**.

[0051] FIG. **2** is a structural diagram of an optical network terminal, in accordance with some embodiments. In order to clearly show a connection relationship between the optical module **200** and the optical network terminal **100**, FIG. **2** only shows portions of the optical network terminal **100** related to the optical module **200**. As shown in FIG. **2**, the optical network terminal **100** further includes a PCB circuit board **105** disposed in the housing, a cage **106** disposed on a surface of the PCB circuit board **105**, a heat sink **107** disposed on the cage **106**, and an electrical connector disposed inside the cage **106**. The electrical connector is configured to connect to the electrical port of the optical module **200**; and the heat sink **107** has protruding portions such as fins that increase a heat dissipation area.

[0052] The optical module **200** is inserted into the cage **106** of the optical network terminal **100** and is fixed by the cage **106**. Heat generated by the optical module **200** is conducted to the cage **106** and is then diffused by the heat sink **107**. After the optical module **200** is inserted into the cage **106**, the electrical port of the optical module **200** is connected to the electrical connector inside the

cage **106**, so that the bidirectional electrical signal connection is established between the optical module **200** and the optical network terminal **100**. In addition, the optical port of the optical module **200** is connected to the optical fiber **101**, so that the bidirectional optical signal connection is established between the optical module **200** and the optical fiber **101**.

[0053] FIG. **3** is a structural diagram of an optical module, in accordance with some embodiments. FIG. **4** is an exploded view of an optical module, in accordance with some embodiments. As shown in FIGS. **3** and **4**, the optical module **200** includes a shell, and a circuit board **300**, a light emitting assembly **400**, and a light receiving assembly **500** that are disposed inside the shell. However, the structure of the optical module **200** is not limited thereto. In some embodiments, the optical module **200** includes one of the light emitting assembly **400** and the light receiving assembly **500**.

[0054] The shell includes an upper shell **201** and a lower shell **202**. The upper shell **201** covers the lower shell **202** to form the shell with two openings **204** and **205**, and an outer contour of the shell is generally in a cuboid shape.

[0055] In some embodiments, the lower shell **202** includes a bottom plate **2021** and two lower side plates **2022** that are arranged perpendicular to the bottom plate **2021** and located on two sides of the bottom plate **2021** respectively; and the upper shell **201** includes a cover plate **2011**, and the cover plate **2011** covers the two lower side plates **2022** of the lower shell **202** to form the shell.

[0056] In some embodiments, the lower shell **202** includes a bottom plate **2021** and two lower side plates **2022** that are arranged perpendicular to the bottom plate **2021** and located on two sides of the bottom plate **2021** respectively; the upper shell **201** includes a cover plate **2011** and two upper side plates that are arranged perpendicular to the cover plate **2011** and located on two sides of the cover plate **2011** respectively. The two upper side plates are combined with the two lower side plates **2022**, so that the upper shell **201** covers the lower shell **202**.

[0057] A direction in which a connecting line between the two openings **204** and **205** extends may or may not be the same as a length direction of the optical module **200**. For example, the opening **204** is located at an end (a right end in FIG. **3**) of the optical module **200**, and the opening **205** is located at an end (a left end in FIG. **3**) of the optical module **200**. Alternatively, the opening **204** is located at an end of the optical module **200**, while the opening **205** is located at a side of the optical module **200**. The opening **204** is the electrical port, and a connecting finger **301** of the circuit board **300** extends from the electrical port **204** and is inserted into the master monitor (e.g., the optical network terminal **100**). The opening **205** is the optical port **205** and is configured to connect to an external optical fiber **101**, so that the optical fiber **101** is connected to the light emitting assembly **400** and the light receiving assembly **500** in the optical module **200**.

[0058] With help of an assembly manner of combining the upper shell **201** with the lower shell **202**, it is helpful to install components such as the circuit board **300**, the light emitting assembly **400**, and the light receiving assembly **500** into the shell, and these components may be encapsulated and protected by the upper shell **201** and the lower shell **202**. In addition, during assembly of the components such as the circuit board **300**, the light emitting assembly **400**, and the light receiving assembly **500**, it is also helpful to arrange positioning elements, heat dissipation elements, and electromagnetic shielding elements of these components, which facilitates automated implementation of production.

[0059] In some embodiments, the upper shell **201** and the lower shell **202** are made of a metallic material, which facilitates electromagnetic shielding and heat dissipation.

[0060] In some embodiments, the optical module **200** further includes an unlocking component **203** located outside the shell thereof, and the unlocking component **203** is configured to implement or release a fixed connection between the optical module **200** and the master monitor.

[0061] For example, the unlocking component **203** is on outer sides of the two lower side plates **2022** of the lower shell **202** and includes an engagement component that is matched with a cage of the master monitor (e.g., the cage **106** of the optical network terminal **100**). When the optical module **200** is inserted into the cage **106**, the optical module **200** is fixed in the cage **106** by the

engagement component of the unlocking component **203**. When the unlocking component **203** is pulled, the engagement component of the unlocking component **203** moves with the pulling, and then a connection relationship between the engagement component and the master monitor is changed to release an engagement relationship between the optical module **200** and the master monitor, and thus the optical module **200** may be drawn out of the cage **106**.

[0062] The circuit board **300** includes circuit wires, electronic elements, and chips. Through the circuit wires, the electronic elements and the chips are connected together according to a circuit design, so as to achieve functions such as power supply, electrical signal transmission, and grounding. The electronic elements may include, for example, a capacitor, a resistor, a triode, and a metal-oxide-semiconductor field-effect transistor (MOSFET). The chips may include, for example, a microcontroller unit (MCU), a laser driver chip, a transimpedance amplifier (TIA), a limiting amplifier, a clock and data recovery (CDR) chip, a power management chip, or a digital signal processing (DSP) chip.

[0063] The circuit board **300** is generally a rigid circuit board, and the rigid circuit board may also achieve a bearing function due to its relatively hard material, for example, the rigid circuit board may stably bear the electronic elements and the chips. The rigid circuit board may also be inserted into the electrical connector inside the cage **106** of the master monitor.

[0064] The circuit board **300** further includes the connecting finger **301** formed on a surface of the end thereof, and the connecting finger **301** is composed of a plurality of pins independent of each other. The circuit board **300** is inserted into the cage **106** and is conductively connected to the electrical connector inside the cage **106** through the connecting finger **301**. The connecting finger **301** may be disposed on only a surface (e.g., an upper surface shown in FIG. 4) of the circuit board **300** or may be disposed on both upper and lower surfaces of the circuit board **300** to adapt to an occasion with a demand for a large number of pins. The connecting finger **301** is configured to establish electrical connection with the master monitor to achieve power supply, grounding, transmission of I2C signals, or transmission of data signals. Of course, flexible circuit boards are also used in some optical modules. A flexible circuit board is generally used in conjunction with a rigid circuit board as a supplement for the rigid circuit board.

[0065] In some embodiments, the light emitting assembly **400** and the light receiving assembly **500** are disposed on a side of the circuit board **300** and away from the connecting finger **301**, and the light emitting assembly **400** and the light receiving assembly **500** are arranged in a stack (e.g., stacked one on top of the other). For example, as shown in FIG. 4, the light emitting assembly **400** is disposed on a side of the light receiving assembly **500** proximate to the upper shell **201**, that is, the light emitting assembly **400** is closer to the upper shell **201** than the light receiving assembly **500**. Alternatively, the light emitting assembly **400** is disposed on a side of the light receiving assembly **500** away from the upper shell **201**, that is, the light emitting assembly **400** is farther away from the upper shell **201** than the light receiving assembly **500**.

[0066] In a case where the light emitting assembly **400** is closer to the upper shell **201** than the light receiving assembly **500**, the lower shell **202** may directly support the light receiving assembly **500**, and the light receiving assembly **500** supports the light emitting assembly **400**. Of course, in some embodiments, the lower shell **202** may further support the light receiving assembly **500** through a cushion block.

[0067] In some embodiments, the light emitting assembly **400** and the light receiving assembly **500** are physically separated from the circuit board **300**, and each are electrically connected to the circuit board **300** through a corresponding flexible circuit board or electrical connector. The volumes of the light emitting assembly **400** and the light receiving assembly **500** each are large, and the light emitting assembly **400** and the light receiving assembly **500** are not capable of disposing on the circuit board **300**. Therefore, the light emitting assembly **400** and the light receiving assembly **500** are separated from the circuit board **300** and the electrical connection is performed through the flexible circuit boards or electrical connectors.

[0068] In some embodiments, the optical module **200** further includes a first fiber optic adapter **600** and a second fiber optic adapter **700**. The first fiber optic adapter **600** is connected to the light emitting assembly **400** and is configured to transmit an optical signal from the light receiving assembly **500** to the outside of the optical module **200**. The second fiber optic adapter **700** is connected to the light receiving assembly **500** and is configured to transmit an optical signal from the outside of the optical module **200** to the light receiving assembly **500**.

[0069] FIG. **5** is a local sectional view of an optical module, in accordance with some embodiments. As shown in FIG. **5**, the light emitting assembly **400** includes a light emitting housing **420**. The first fiber optic adapter **600** (i.e., the fiber optic adapter) is disposed at an end of the light emitting housing **420** away from the circuit board **300**. The light receiving assembly **500** includes a light receiving housing **520**, and the second fiber optic adapter **700** is disposed at an end of the light receiving housing **520** away from the circuit board **300**. The circuit board **300** is electrically connected with the light emitting assembly **400** and the light receiving assembly **500** through the flexible circuit boards.

[0070] It can be understood that a size of an overall shape of the optical module **200** needs to be matched with a size of an interface of the master monitor. However, the volumes of the light emitting assembly **400** and the light receiving assembly **500** are large, so that the light emitting assembly **400** and the light receiving assembly **500** are not capable of disposing on the circuit board **300**. Therefore, the circuit board **300** is set to be separated from the light emitting assembly **400** and the light receiving assembly **500**, and a connection between the circuit board **300** and the light emitting assembly **400** and a connection between the circuit board **300** and the light receiving assembly **500** are implemented through the flexible circuit boards.

[0071] In some embodiments, as shown in FIG. **5**, in a thickness direction of the optical module **200** (i.e., the XY direction in FIG. **5**), the first fiber optic adapter **600** and the second fiber optic adapter **700** are located at a same height. The first fiber optic adapter **600** and the second fiber optic adapter **700** are connected to external fiber optic connectors of the optical module **200**.

[0072] FIG. **6** is an exploded view of a light emitting assembly and a light receiving assembly in an optical module, in accordance with some embodiments. As shown in FIG. **6**, the light receiving assembly **500** further includes a light receiving cover plate **530**, and the light receiving cover plate **530** is configured to cover the light receiving housing **520**. The light receiving assembly **500** includes a lens group, a light receiving chip, a trans-impedance amplifier, and other elements related to light receiving. The light receiving housing **520** receives an optical signal from the outside of the optical module **200** through the second fiber optic adapter **700** and transmits the received optical signal to the light receiving chip through lenses and other optical elements disposed in the light receiving housing **520**.

[0073] In some embodiments, the light receiving assembly **500** further includes a second flexible circuit board **510**, and the light receiving housing **520** includes a second inserting hole **521**. The second inserting hole **521** is disposed on a side wall of the light receiving housing **520** away from the second fiber optic adapter **700**. An end of the second flexible circuit board **510** is connected with the light receiving housing **520** through the second inserting hole **521** and is electrically connected with the electrical elements such as the light receiving chip and the trans-impedance amplifier. Another end of the second flexible circuit board **510** is electrically connected with the circuit board **300**.

[0074] FIG. **7** is an exploded view of a light emitting assembly in an optical module, in accordance with some embodiments. FIG. **8** is a diagram showing a partial structure of a light emitting assembly in an optical module, in accordance with some embodiments. As shown in FIGS. **7** and **8**, the light emitting assembly **400** further includes a first flexible circuit board **410** (seeing FIG. **20**), a light emitting cover plate **430** and a light emitting component **440**. The light emitting cover plate **430** is configured to cover the light emitting housing **420**. The light emitting cover plate **430** and the light emitting housing **420** jointly define a light emitting cavity **403**. The light emitting

component **440** is disposed in the light emitting cavity **403**. The light emitting component **440** includes a laser **441**, a light detector **480**, a laser driver **401**, and a lens group.

[0075] The light emitting component **440** is configured to emit and converge a plurality of light beams carrying signals (i.e., optical signals), after multiplexing the plurality of light beams carrying signals into one optical signal (i.e., a merge composite optical signal), and the light-emitting component **440** is electrically connected to the first fiber optic adapter **600**, so that the plurality of light beams are emitted through one optical fiber. An end of the first flexible circuit board **410** is electrically connected with the laser **441**, the optical detector **480**, and the laser driver **401** through a ceramic adapter block, and another end of the first flexible circuit board **410** is electrically connected with the circuit board **300**. In some embodiments, the light emitting housing **420** and the light emitting cover plate **430** each are metal material structural members, such as die-casting and milling metal members.

[0076] In some embodiments, as shown in FIGS. **7** and **8**, the light emitting component **440** includes a plurality of lasers **441**, a plurality of collimating lenses **442**, a plurality of wavelength division multiplexers, and the lens group. The plurality of lasers **441** are configured to emit the plurality of light beams (i.e., optical signals) with different wavelengths.

[0077] For example, one laser **441** emits one optical signal. The plurality of collimating lenses **442** correspond to the plurality of lasers **441** and are arranged in light-exit directions of the plurality of lasers **441**. The collimating lens **442** is configured to collimate the light beam emitted by the corresponding laser **441**.

[0078] The wavelength division multiplexer is disposed in light-exit directions of a part of the plurality of collimating lenses **442** and is configured to multiplex the optical signals with different wavelengths into one composite optical signal. The lens group is disposed in light-exit directions of the plurality of wavelength division multiplexers and is configured to couple the composite optical signals transmitted by the plurality of wavelength division multiplexers into the first fiber optic adapter **600** after reflection and/or refraction, so as to achieve the light-emitting.

[0079] FIG. **9A** is a top view showing a portion of structure of a light emitting assembly in an optical module, in accordance with some embodiments. FIG. **9B** is an optical path diagram of a light emitting assembly in an optical module, in accordance with some embodiments. As shown in FIGS. **9A** and **9B**, the light emitting component **440** includes eight lasers **441** and eight collimating lenses **442**, the plurality of wavelength division multiplexers include a first wavelength division multiplexer **443** and a second wavelength division multiplexer **444**, and the lens group includes a first lens **445**, a second lens **446**, and a third lens **447**.

[0080] When the light emitting assembly **400** is operating, the eight lasers **441** emit eight optical signals with different wavelengths. The eight collimating lenses **442** correspond to the eight lasers **441** and are arranged in the light-exit directions of the eight lasers **441** respectively, so as to collimate the eight optical signals with different wavelengths.

[0081] The first wavelength division multiplexer **443** corresponds to four collimating lenses **442** of the eight collimating lenses **442** adjacent to each other and is disposed in light-exit directions of the four collimating lenses **442**. The second wavelength division multiplexer **444** corresponds to the other four collimating lenses **442** of the eight collimating lenses **442** adjacent to each other and is disposed in light-exit directions of the other four collimating lenses **442**. In this way, four collimated optical signals may be incident to the first wavelength division multiplexer **443**, and the other four collimated optical signals may be incident into the second wavelength division multiplexer **444**.

[0082] The four collimated optical signals entering the first wavelength division multiplexer **443** may be reflected in the first wavelength division multiplexer **443** and multiplexed into a first composite optical signal, which exits from the first wavelength division multiplexer **443**. Similarly, the four collimated optical signals entering the second wavelength division multiplexer **444** may be reflected in the second wavelength division multiplexer **444** and multiplexed into the second

composite optical signal, which exits from the second wavelength division multiplexer **444**.
[0083] The first lens **445** is arranged in a light-exit direction of the first wavelength division multiplexer **443**, and the second lens **446** is arranged in a light-exit direction of the first lens **445**. In this way, the first composite optical signal exited from the first wavelength division multiplexer **443** may sequentially transmit through the first lens **445** and the second lens **446**.
[0084] The third lens **447** is arranged in a light-exit direction of the second wavelength division multiplexer **444**. The second composite optical signal exited from the second wavelength division multiplexer **444** is incident to the third lens **447** and reflected from the third lens **447** to the second lens **446**, then reflected from the second lens **446** to the first lens **445**, and then reflected from the first lens **445** to the second lens **446**, and transmitted through the second lens **446**. In this way, the first composite optical signal and the second composite optical signal entering the second lens **446** may be combined in the second lens **446**, that is, the first composite optical signal and the second composite optical signal may be combined into the merge composite optical signal in the second lens **446**, and the merge composite optical signal may be converged and coupled into the first fiber optic adapter **600**.
[0085] In some embodiments, as shown in FIG. 8, the first wavelength division multiplexer **443** includes a glass carrier **4431** and a plurality of narrowband filters **4432**. The plurality of narrowband filters **4432** are disposed on a side of the glass carrier **4431** proximate to the lasers **441**. It will be noted that, in practical applications, the number of narrowband filters **4432** may be increased or decreased, so as to adjust the beam splitting band corresponding to the first wavelength division multiplexer **443**.
[0086] For example, the first wavelength division multiplexer **443** includes four narrowband filters **4432**. The four narrowband filters **4432** are used to selectively filter the optical signals entering the first wavelength division multiplexer **443**, so that four light beams satisfying the requirements may enter the first wavelength division multiplexer **443**.
[0087] A portion of surface of the glass carrier **4431** is coated with a reflective film. For example, a surface of the glass carrier **4431** proximate to the collimating lenses **442** and a surface of the glass carrier **4431** away from the collimating lenses **442** each are coated with the reflective film, so that the optical signals entering the first wavelength division multiplexer **443** may be reflected in the glass carrier **4431** coated with the reflective film.
[0088] For example, a first optical signal is reflected in the glass carrier **4431** and combined with a second optical signal, so as to form a first combined light. The first combined light continues to reflect in the glass carrier **4431** coated with the reflective film, and combines with a third optical signal, so as to form a second combined light. Then, the second combined light continues to reflect in the glass carrier **4431** coated with the reflective film and combines with a fourth optical signal, so as to form the first composite optical signal. The first composite optical signal exits from a light exiting surface of the first wavelength division multiplexer **443**.
[0089] Therefore, the first wavelength division multiplexer **443** combines four collimated optical signals into the first composite optical signal, which exits through the light exiting surface of the first wavelength division multiplexer **443**.
[0090] In some embodiments, as shown in FIG. 8, the glass carrier **4431** includes a light incident surface. The light incident surface is a side surface of the glass carrier **4431** proximate to the collimating lenses **442**. The light incident surface is disposed obliquely, that is, an end of the light incident surface is closer to the collimating lenses **442** than another end of the light incident surface.
[0091] For example, referring to FIG. 7, an end of the light incident surface proximate to the second wavelength division multiplexer **444** is closer to the collimating lenses **442** than an end of the light incident surface away from the second wavelength division multiplexer **444**.
[0092] The first wavelength division multiplexer **443** further includes an incident optical port, and the incident optical port is disposed on the light incident surface of the first wavelength division

multiplexer **443**, that is, the incident optical port is disposed on the side of the glass carrier **4431** proximate to the collimating lenses **442**. The incident optical port is used to receive the collimated optical signals exited from the collimating lenses **442**. It will be noted that the first wavelength division multiplexer **443** includes one or more incident optical ports.

[0093] For example, the first wavelength division multiplexer **443** includes four incident optical ports. One incident optical port receives a collimated optical signal exited from one collimating lens **442** and transmits the received collimated optical signal into the glass carrier **4431** for reflection and combining.

[0094] It can be understood that, when a collimated optical signal exited from a collimating lens **442** is incident on the incident optical port, a part of the collimated optical signal may enter the first wavelength division multiplexer **443** through the incident optical port, and another part of the collimated optical signal may be reflected at the incident optical port. In this case, since the light incident surface is disposed obliquely, the another part of the collimated optical signal will not be reflected to return to the collimating lens **442** along its incident direction, thereby avoiding affecting the collimating performance of the collimating lenses **442**.

[0095] It will be noted that, the second wavelength division multiplexer **444** has the same structure and function as that of the first wavelength division multiplexer **443**. The second wavelength division multiplexer **444** combines the other four collimated optical signals into the second composite optical signal, which exits from a light exiting surface of the second wavelength division multiplexer **444**, and details will not be repeated herein.

[0096] FIG. **9C** is an optical path diagram of a lens group in an optical module, in accordance with some embodiments. As shown in FIG. **9C**, the first lens **445** includes an exit surface **4451**, the second lens **446** includes a first incident surface **4461**, and the third lens **447** includes a second incident surface **4471**.

[0097] The first composite optical signal exited from the first wavelength division multiplexer **443** passes through the first lens **445**, then exits from the exit surface **4451** and is incident on the first incident surface **4461**, and then transmits through the second lens **446**.

[0098] The second composite optical signal exited from the second wavelength division multiplexer **444** is incident on the second incident surface **4471** of the third lens **447**. The second incident surface **4471** is provided with a reflective film. Therefore, the second composite optical signal will be reflected at the reflective film of the second incident surface **4471** after being incident on the second incident surface **4471**, so that the second composite optical signal is reflected from the third lens **447** to the first incident surface **4461** of the second lens **446**. Then, the second composite optical signal is reflected from the first incident surface **4461** to the exit surface **4451** of the first lens **445**, then reflected to the first incident surface **4461** of the second lens **446**, and then transmitted through the second lens **446**.

[0099] In some embodiments, the first incident surface **4461** of the second lens **446** is provided with an angle selection film, so as to reflect the second composite optical signal when incident on the first incident surface **4461** for a first time. For example, the angle selection film selectively transmits the optical signals incident on the first incident surface **4461**, in other words, the angle selection film allows optical signals incident at a preset angle to transmit through.

[0100] For example, in a case where an incident angle of an optical signal incident on the first incident surface **4461** satisfies the preset angle, the optical signal may transmit through the second lens **446**. In a case where an incident angle of an optical signal incident on the first incident surface **4461** does not satisfy the preset angle, the optical signal is reflected at the first incident surface **4461**.

[0101] In some embodiments, when the first composite optical signal exited from the first lens **445** is incident on the first incident surface **4461** of the second lens **446**, an incident angle of the first composite optical signal satisfies the preset angle, therefore, the first composite optical signal may transmit through the second lens **446**. An incident angle of the second composite optical signal

reflected from the third lens **447** to the first incident surface **4461** is greater than the preset angle, therefore, the second composite optical signal is reflected at the first incident surface **4461** (e.g., reflected to the first lens **445**).

[0102] For example, the preset angle that the angle selection film allows light beams to pass through is in a range of 0° to 45° inclusive. The incident angle of the first composite optical signal passing through the first lens **445** is α (for example, the α is 37°) when the first composite optical signal is incident on the first incident surface **4461**, and the incident angle satisfies the preset angle that the angle selection film allows light beams to pass through. Therefore, the first composite optical signal may transmit through the angle selection film. The incident angle of the second multiplexed optical signal reflected from the third lens **447** to the first incident surface **4461** is β (for example, the β is 57°), which exceeds the preset angle that the angle selection film allows light beams to pass through. Therefore, the second composite optical signal is reflected when incident on the first incident surface **4461**.

[0103] Generally, the lenses in the lens group transmit or reflect the optical signals with different wavelengths by using coating, so as to achieve the combination of the plurality of optical signals. When the lens group in the present disclosure combines the plurality of optical signals, the plurality of optical signals are further selectively transmitted or reflected according to the incident angles of the plurality of optical signals when incidenting on the lenses, so that the combination may be achieved when the wavelengths interval between the plurality of optical signals are small.

[0104] It will be noted that some embodiments of the present disclosure provide that the first incident surface **4461** of the second lens **446** is not limited to being provided with the angle selection film but may be provided with other films. In a case where the first incident surface **4461** is provided with other films, they are within the scope of the present disclosure as long as the films can achieve that the first composite optical signal is capable of transmitting through the first incident surface **4461** and the second composite optical signal is capable of reflecting at the first incident surface **4461**.

[0105] It can be understood that, in order to enable the second composite optical signal to pass through the second lens **446**, the incident angle when the second composite optical signal is incident on the first incident surface **4461** needs to be reduced. Therefore, by setting relative positional relationship and angle relationship between the first lens **445**, the second lens **446**, and the third lens **447**, the incident angle of the second composite optical signal satisfies the preset angle that the angle selection film allows light beams to pass through, when the second composite optical signal is reflected from the first incident surface **4461** to the exit surface **4451** of the first lens **445**, and then reflected to the first incident surface **4461**.

[0106] In some embodiments, the exit surface **4451** of the first lens **445** is provided with an optical filter. The optical filter may select a wavelength of an incident light beam, that is, a light beam with a wavelength within a preset wavelength range may pass through the optical filter, and a light beam with a wavelength outside the preset wavelength range will reflect when incident on the optical filter.

[0107] Wavelengths of the first composite optical signal satisfy the preset wavelength range, therefore, the first composite optical signal may pass through the exit surface **4451** of the first lens **445**. A wavelength of the second composite optical signal does not satisfy the preset wavelength range of the optical filter, therefore, the second composite optical signal reflected from the second lens **446** to the exit surface **4451** of the first lens **445** is reflected when incident on the optical filter, so that the second composite optical signal is reflected from the exit surface **4451** of the first lens **445** to the first incident surface **4461** of the second lens **446**.

[0108] It can be understood that, the second composite optical signal is incident on the first incident surface **4461** of the second lens **446** again, and the incident angle of the second composite optical signal satisfies the preset angle that the angle selection film allows light beams to pass through after the reflection of the first incident surface **4461** of the second lens **446** and the

reflection of the exit surface **4451** of the first lens **445**, so that the second composite optical signal may pass through the second lens **446**.

[0109] As shown in FIG. **9C**, the first composite optical signal is combined with the second composite optical signal in the second lens **446**, that is, the first composite optical signal and the second composite optical signal are combined into the merge composite optical signal in the second lens **446**, and the merge composite optical signal exits from the second lens **446** and is then coupled to the first fiber optic adapter **600**.

[0110] In some embodiments, as shown in FIG. **9B**, the light emitting assembly **400** further includes a converging lens **402**. The converging lens **402** is disposed between the second lens **446** and an end of an optical fiber plug of the first fiber optic adapter **600** and is located on an optical path of the merge composite optical signal. In this way, the merge composite optical signal exited from the second lens **446** is converged by the converging lens **402** and then coupled to the first fiber optic adapter **600**, thereby improving the coupling efficiency.

[0111] In some embodiments, as shown in FIG. **7** and FIGS. **9A** to **9C**, the light emitting component **440** further includes a flat lens **448**. The flat lens **448** is disposed in a side wall of the light emitting housing **420** proximate to the first fiber optic adapter **600** and located between the second lens **446** and the converging lens **402** and located on the optical path of the merge composite optical signal. The flat lens **448** is configured to transmit optical signals between the light emitting component **440** and the first fiber optic adapter **600**.

[0112] For example, referring to FIG. **22**, the flat lens **448** is a transparent glass sheet coated with an anti-reflective coating corresponding to a certain wavelength, and is disposed obliquely with respect to the merge composite optical signal (i.e., an incident angle of the merge composite optical signal incident on the flat lens **448** is less than or equal to 8°), so that a transmissivity of the flat lens **448** may be improved, so as to prevent the light reflection from affecting the performance of the light emitting component **440**.

[0113] The light emitting component **440** provided by some embodiments of the present disclosure includes the eight lasers **441** and the eight collimating lenses **442**, the plurality of wavelength division multiplexers include the first wavelength division multiplexer **443** and the second wavelength division multiplexer **444**, and the lens group includes the first lens **445**, the second lens **446**, and the third lens **447**.

[0114] The eight lasers **441** emit the eight optical signals. The eight collimating lenses **442** are respectively arranged in the light-exit directions of the eight lasers **441**, so as to collimate the eight optical signals to obtain eight collimated optical signals. The first wavelength division multiplexer **443** and the second wavelength division multiplexer **444** are disposed in the light-exit directions of the eight collimating lenses **442**. The first wavelength division multiplexer **443** and the second wavelength division multiplexer **444** each include four input channels. Four of the eight collimated optical signals enter the four channels of the first wavelength division multiplexer **443**, and the other four collimated optical signals enter the four channels of the second wavelength division multiplexer **444**. The first wavelength division multiplexer **443** converts the four collimated optical signals into the first composite optical signal, and the second wavelength division multiplexer **444** converts the four collimated optical signal into the second composite optical signal.

[0115] The first lens **445** is arranged in the light-exit direction of the first wavelength division multiplexer **443**, and the second lens **446** is arranged in the light-exit direction of the first lens **445**. The first composite optical signal exited from the first wavelength division multiplexer **443** transmits through the first lens **445** and the second lens **446** directly.

[0116] The third lens **447** is arranged in the light-exit direction of the second wavelength division multiplexer **444**. The second composite optical signal exited from the second wavelength division multiplexer **444** is reflected by the third lens **447** to the second lens **446**. Since the incident angle of the reflected second composite optical signal does not satisfy the preset angle that the second lens **446** allows light beams to pass through, the reflected second composite optical signal is reflected at

the first incident surface **4461** of the second lens **446** and reflected to the exit surface **4451** of the first lens **445**.

[0117] The reflected second composite optical signal is reflected on the exit surface **4451** of the first lens **445**, and the reflected second composite optical signal is incident on the first incident surface **4461** of the second lens **446** again. Since the incident angle of the reflected second composite optical signal is reduced, the second composite optical signal reflected from the exit surface **4451** of the first lens **445** to the second lens **446** may transmit through the second lens **446** directly.

[0118] The first composite optical signal and the second composite optical signal entering the second lens **446** after reflection may be combined into the merge composite optical signal in the second lens **446**. The merge composite optical signal is coupled to the first fiber optic adapter **600** after passing through the flat lens **448** and the focusing lens **402**, thereby implementing the transmission of the eight wavelength division multiplexing optical signals.

[0119] The optical module **200** provided by some embodiments of the present disclosure multiplexes eight optical signals into two composite optical signals (i.e., the first wavelength division multiplexer **443** and the second wavelength division multiplexer **444**) through two wavelength division multiplexers, and then the propagation directions of the two composite optical signals are adjusted through three lenses, so as to combine the two composite optical signals into the merge composite optical signal. The merge composite optical signal is finally converged and coupled to the external optical fiber. In this way, it is conducive to reducing the volume of the light emitting assembly **400**, making the light emitting assembly **400** occupy less space in the optical module **200**, and implementing a miniaturization design of the optical module **200**.

[0120] The light emitting component **440** of the present disclosure is not limited thereto. In some other embodiments, the light emitting component **440** includes a plurality of lasers **441**, a plurality of collimating lenses **442**, and a lens group. The main difference between the embodiments lies in the structure of the lens group.

[0121] The lens group is arranged in light-exit directions of the plurality of collimating lenses **442** and is configured to multiplex a plurality of optical signals with different wavelengths into a merge composite optical signal after reflection and/or refraction and couple the merge composite optical signal to the first fiber optic adapter **600**, so as to implement the light-emitting.

[0122] FIG. **10A** is an optical path diagram of another light emitting assembly in an optical module, in accordance with some embodiments. FIG. **10B** is an optical path diagram of another lens group in an optical module, in accordance with some embodiments.

[0123] In some embodiments, as shown in FIGS. **10A** and **10B**, the light emitting component **440** includes eight lasers **441**, eight collimating lenses **442**, and the lens group.

[0124] When the light emitting assembly **400** is operating, the eight lasers **441** emit eight optical signals with different wavelengths. The eight collimating lenses **442** correspond to the eight lasers **441** and are arranged in the light-exit directions of the eight lasers **441** respectively, so as to collimate the eight optical signals with different wavelengths.

[0125] The eight lasers **441** include a first laser **441A**, a second laser **441B**, a third laser **441C**, a fourth laser **441D**, a fifth laser **441E**, a sixth laser **441F**, a seventh laser **441G**, and an eighth laser **441H**. The first laser **441A** to the eighth laser **441H** are arranged at an interval along a width direction of the light emitting housing **420** (i.e., the S direction in FIG. **10A**).

[0126] The eight collimating lenses **442** include a first collimating lens **4421**, a second collimating lens **4422**, a third collimating lens **4423**, a fourth collimating lens **4424**, a fifth collimating lens **4425**, a sixth collimating lens **4426**, a seventh collimating lens **4427**, and an eighth collimating lens **4428**. The first collimating lens **4421** to the eighth collimating lens **4428** are arranged corresponding to the first laser **441A** to the eighth laser **441H** respectively.

[0127] The lens group includes a fourth lens **4401**, a fifth lens **4402**, a sixth lens **4403**, a seventh lens **4404**, an eighth lens **4405**, a ninth lens **4406**, a tenth lens **4407**, a first polarizer **4408**, a second

polarizer **4409**, and a polarizing beam member **4419**. The polarizing beam member **4419** includes a first lozenge lens **4410** and a polarizing beam splitter (PBS) **4411**.

[0128] It will be noted that, for ease of description, the obliquely disposing described below is that a lens or a surface of the lens group is disposed obliquely with the S direction in FIG. **10A** or the light-exit directions of the lasers **441**. In other words, when the lens or the surface of the lens group is disposed obliquely, the lens or the surface is not perpendicular to or parallel to the S direction in FIG. **10A**, nor perpendicular to or parallel to the light-exit directions of the lasers **441**, and the lens or the surface is perpendicular to the bottom plate of the light emitting housing **420**.

[0129] The fourth lens **4401** is disposed obliquely on light-exit paths of the first collimating lens **4421** and the second collimating lens **4422**, and a surface of the fourth lens **4401** facing toward the first collimating lens **4421** and the second collimating lens **4422** is provided with a reflective film.

[0130] The fifth lens **4402** is disposed obliquely in an intersection of a light-exit direction of the fourth lens **4401** and a light-exit direction of the third collimating lens **4423**. The seventh lens **4404** is disposed obliquely in an intersection of a light-exit direction of the fifth lens **4402** and a light-exit direction of the fifth collimating lens **4425**. The ninth lens **4406** is disposed obliquely in an intersection of a light-exit direction of the seventh lens **4404** and a light-exit direction of the seventh collimating lens **4427**. The first polarizer **4408** is disposed in a light-exit direction of the ninth lens **4406**.

[0131] The surfaces of the fifth lens **4402**, the seventh lens **4404**, and the ninth lens **4406** facing toward the plurality of collimating lenses **442** each are provided with the optical filter, so that the fifth lens **4402**, the seventh lens **4404**, and the ninth lens **4406** may selectively transmit or reflect the optical signals with different wavelengths.

[0132] For example, the fifth lens **4402** may transmit an optical signal exited from the first collimating lens **4421** and reflect an optical signal exited from the third collimating lens **4423**. The seventh lens **4404** may transmit the optical signals exited from the first collimating lens **4421** and the third collimating lens **4423** and reflect an optical signal exited from the fifth collimating lens **4425**. The ninth lens **4406** may transmit an optical signal exited from the seventh collimating lens **4427** and reflect the optical signals exited from the first collimating lens **4421**, the third collimating lens **4423**, and the fifth collimating lens **4425**.

[0133] In this way, a first collimated optical signal (i.e., a first optical signal emitted by the first laser **441A**) exited from the first collimating lens **4421** is reflected by the fourth lens **4401**, and then transmits through the fifth lens **4402** and the seventh lens **4404**, and is reflected to the first polarizer **4408** by the ninth lens **4406**.

[0134] A third collimated optical signal (i.e., a third optical signal emitted by the third laser **441C**) exited from the third collimating lens **4423** is reflected when incident on the fifth lens **4402** and is multiplexed with the first collimated optical signal into a first sub composite optical signal. The first sub composite optical signal is reflected at the ninth lens **4406** after transmitting through the seventh lens **4404**.

[0135] A fifth collimated optical signal (i.e., a fifth optical signal emitted by the fifth laser **441E**) exited from the fifth collimating lens **4425** is reflected when incident on the seventh lens **4404** and is multiplexed with the first sub composite optical signal into a second sub composite optical signal. The second sub composite optical signal is reflected at the ninth lens **4406**.

[0136] A seventh collimated optical signal (i.e., the seventh optical signal emitted by the seventh laser **441G**) exited from the seventh collimating lens **4427** transmits through the ninth lens **4406** and is multiplexed with the second sub composite optical signal into a first composite optical signal. The first composite optical signal exits from the ninth lens **4406** and is incident on the first polarizer **4408**.

[0137] The sixth lens **4403** is disposed obliquely in an intersection of a light-exit direction of the fourth lens **4401** and a light-exit direction of the fourth collimating lens **4424**. The eighth lens **4405** is disposed obliquely in an intersection of a light-exit direction of the sixth lens **4403** and a light-

exit direction of the sixth collimating lens **4426**. The tenth lens **4407** is disposed obliquely in an intersection of a light-exit direction of the eighth lens **4405** and a light-exit direction of the eighth collimating lens **4428**. The second polarizer **4409** is disposed in a light-exit direction of the tenth lens **4407**.

[0138] Similarly, the surfaces of the sixth lens **4403**, the eighth lens **4405**, and the tenth lens **4407** facing toward the plurality of collimating lenses **442** each are provided with the optical filter, so that the sixth lens **4403**, the eighth lens **4405**, and the tenth lens **4407** may selectively transmit or reflect the optical signals with different wavelengths.

[0139] For example, the sixth lens **4403** may transmit an optical signal exited from the second collimating lens **4422** and reflect an optical signal exited from the fourth collimating lens **4424**. The eighth lens **4405** may transmit the optical signals exited from the second collimating lens **4422** and the fourth collimating lens **4424** and reflect an optical signal exited from the sixth collimating lens **4426**. The tenth lens **4407** may transmit an optical signal exited from the eighth collimating lens **4428** and reflect the optical signals exited from the second collimating lens **4422**, the fourth collimating lens **4424** and the sixth collimating lens **4426**.

[0140] In this way, a second collimated optical signal (i.e., a second optical signal emitted by the second laser **441B**) exited from the second collimating lens **4422** is reflected by the fourth lens **4401**, and then sequentially transmits through the sixth lens **4403** and the eighth lens **4405** and is reflected to the second polarizer **4409** by the tenth lens **4407**.

[0141] A fourth collimated optical signal (i.e., a fourth optical signal emitted by the fourth laser **441D**) exited from the third collimating lens **4423** is reflected when incident on the sixth lens **4403** and is multiplexed with the second collimated optical signal into a third sub composite optical signal. The third sub composite optical signal is reflected by the tenth lens **4407** after transmitting through the eighth lens **4405**.

[0142] A sixth collimated optical signal (i.e., a sixth optical signal emitted by the sixth laser **441F**) exited from the sixth collimating lens **4426** is reflected when incident on the eighth lens **4405** and is multiplexed with the third sub composite optical signal into a fourth sub composite optical signal. The fourth sub composite optical signal is reflected by the tenth lens **4407**.

[0143] An eighth collimated optical signal (i.e., an eighth optical signal emitted by the eighth laser **441H**) exited from the eighth collimating lens **4428** transmits through the tenth lens **4407** and is multiplexed with the fourth sub composite optical signal into a second composite optical signal. The second composite optical signal exits from the tenth lens **4407** and is incident on the second polarizer **4409**.

[0144] The first polarizer **4408** is configured to convert an optical signal incident on the first polarizer **4408** into a polarized light, and a polarization direction of the polarized light is a first direction. The second polarizer **4409** is configured to convert an optical signal incident on the second polarizer **4409** into a polarized light, and a polarization direction of the polarized light is a second direction. The first direction is perpendicular to the second direction.

[0145] In this way, the first composite optical signal is converted into a polarized light through the first polarizer **4408**, and a polarization direction of the first composite optical signal is the first direction. The second composite optical signal is converted into a polarized light through the second polarizer **4409**, and a polarization direction of the second composite optical signal is the second direction.

[0146] Referring to FIGS. **10A** and **10B**, the first lozenge lens **4410** includes a first surface **44101** and a second surface **44102**. The first surface **44101** is opposite to the second surface **44102**, and a surface of the first lozenge lens **4410** away from the polarizing beam splitter **4411** is the first surface **44101**. The polarizing beam splitter **4411** is located on a light-exit path of the first polarizer **4408**, the first surface **44101** is located on a light-exit path of the second polarizer **4409**, and the first surface **44101** is provided with a reflective film. The polarizing beam splitter **4411** and the first lozenge lens **4410** are arranged in the S direction. The polarizing beam splitter **4411** is connected

with the first lozenge lens **4410**. The polarizing beam splitter **4411** includes a splitter slope **4412**. The splitter slope **4412** is disposed obliquely and connected with the second surface **44102**. The splitter slope **4412** is configured to transmit the polarized light whose polarization direction is the first direction and reflect the polarized light whose polarization direction is the second direction. [0147] The first surface **44101** of the first lozenge lens **4410** is parallel to the splitter slope **4412**. The splitter slope **4412** is located in a light-exit direction of the reflective film of the first lozenge lens **4410**.

[0148] In this way, the first composite optical signal exited from the first polarizer **4408** may transmit through the polarizing beam member **4419** (e.g., the splitter slope **4412**). The second composite optical signal exited from the second polarizer **4409** is reflected to the splitter slope **4412** through the reflective film of the first lozenge lens **4410** and multiplexed with the first composite optical signal into a merge composite optical signal after being reflected by the splitter slope **4412**. The merge composite optical signal is coupled to the first fiber optic adapter **600**, so as to implement the light-emitting.

[0149] It will be noted that, oblique directions of the fourth lens **4401**, the fifth lens **4402**, the sixth lens **4403**, the seventh lens **4404**, the eighth lens **4405**, the ninth lens **4406**, and the tenth lens **4407** are the same.

[0150] For example, the fourth lens **4401**, the fifth lens **4402**, the seventh lens **4404**, and the ninth lens **4406** are parallel to each other, and the fourth lens **4401**, the sixth lens **4403**, the eighth lens **4405**, and the tenth lens **4407** are parallel to each other. The fifth lens **4402** is flush with the sixth lens **4403**, the seventh lens **4404** is flush with the eighth lens **4405**, and the ninth lens **4406** is flush with the tenth lens **4407**.

[0151] An oblique direction of the splitter slope **4412** is the same as that of the surface (i.e., the first surface **44101**) of the first lozenge lens **4410** away from the polarizing beam splitter **4411**. For example, the splitter slope **4412** is parallel to the surface of the first lozenge lens **4410** away from the polarizing beam splitter **4411**.

[0152] FIG. **11A** is an optical path diagram of yet another light emitting assembly in an optical module, in accordance with some embodiments. FIG. **11B** is an optical path diagram of yet another lens group in an optical module, in accordance with some embodiments.

[0153] In some embodiments, as shown in FIGS. **11A** and **11B**, the fourth lens **4401**, the fifth lens **4402**, and the sixth lens **4403** may be configured as a second lozenge lens **4413**, and the seventh lens **4404**, the eighth lens **4405**, the ninth lens **4406**, and the tenth lens **4407** may be configured as a third lozenge lens **4414**. That is, in yet some other embodiments, the lens group includes the second lozenge lens **4413**, the third lozenge lens **4414**, the first polarizer **4408**, the second polarizer **4409**, and the polarizing beam member **4419**, and the polarizing beam member **4419** includes the first lozenge lens **4410** and the polarizing beam splitter **4411**.

[0154] The second lozenge lens **4413** and the third lozenge lens **4414** are arranged along the width direction of the light emitting housing **420** (i.e., the S direction shown in FIG. **11A**).

[0155] The second lozenge lens **4413** includes a reflecting surface **4415** and a first transmissive-reflective surface **4416**, and the third lozenge lens **4414** includes a second transmissive-reflective surface **4417** and a third transmissive-reflective surface **4418**.

[0156] The reflecting surface **4415** is a surface of the second lozenge lens **4413** away from the third lozenge lens **4414**. The reflecting surface **4415** is disposed obliquely, and the reflecting surface **4415** is located on the light-exit paths of the first collimating lens **4421** and the second collimating lens **4422**.

[0157] The first transmissive-reflective surface **4416** is a surface of the second lozenge lens **4413** proximate to the third lozenge lens **4414**. The first transmissive-reflective surface **4416** is disposed obliquely. The first transmissive-reflective surface **4416** is located on the light-exit paths of the third collimating lens **4423** and the fourth collimating lens **4424** and is located on light-exit paths of the reflecting surface **4415**.

[0158] The second transmissive-reflective surface **4417** is a surface of the third lozenge lens **4414** proximate to the second lozenge lens **4413**. The second transmissive-reflective surface **4417** is disposed obliquely. The second transmissive-reflective surface **4417** is located on the light-exit paths of the fifth collimating lens **4425** and the sixth collimating lens **4426** and is located on light-exit paths of the first transmissive-reflective surface **4416**.

[0159] The third transmissive-reflective surface **4418** is a surface of the third lozenge lens **4414** away from the second lozenge lens **4413**. The third transmissive-reflective surface **4418** is disposed obliquely. The third transmissive-reflective surface **4418** is located on the light-exit paths of the seventh collimating lens **4427** and the eighth collimating lens **4428** and is located on light-exit paths of the second transmissive-reflective surface **4417**. The first polarizer **4408** and the second polarizer **4409** are located on the light-exit paths of the third transmissive-reflective surface **4418**.

[0160] The first transmissive-reflective surface **4416**, the second transmissive-reflective surface **4417**, and the third transmissive-reflective surface **4418** each are all provided with the optical filter, so as to selectively transmit or reflect the optical signals with different wavelengths.

[0161] For example, the first transmissive-reflective surface **4416** may transmit the optical signals exited from the first collimating lens **4421** and the second collimating lens **4422** and reflect the optical signals exited from the third collimating lens **4423** and the fourth collimating lens **4424**.

[0162] The second transmissive-reflective surface **4417** may transmit the optical signals exited from the first collimating lens **4421**, the second collimating lens **4422**, the third collimating lens **4423**, and the fourth collimating lens **4424** and reflect the optical signals exited from the fifth collimating lens **4425** and the sixth collimating lens **4426**.

[0163] The third transmissive-reflective surface **4418** may transmit the optical signals exited from the seventh collimating lens **4427** and the eighth collimating lens **4428**, and reflect the optical signals exited from the first collimating lens **4421**, the second collimating lens **4422**, the third collimating lens **4423**, the fourth collimating lens **4424**, the fifth collimating lens **4425**, and the sixth collimating lens **4426**.

[0164] In this way, the first collimated optical signal and the second collimated optical signal exited from the first collimating lens **4421** and the second collimating lens **4422** are respectively reflected by the reflecting surface **4415**, transmitted through the first transmissive-reflective surface **4416** and the second transmissive-reflective surface **4417**, and reflected at the third transmissive-reflective surface **4418**.

[0165] The third collimated optical signal and the fourth collimated optical signal exited from the third collimating lens **4423** and the fourth collimating lens **4424** are reflected when incident on the first transmissive-reflective surface **4416** and are respectively multiplexed with the first collimated optical signal and the second collimated optical signal into the first sub composite optical signal and the third sub composite optical signal. The first sub composite optical signal and the third sub composite optical signal each transmit through the second transmissive-reflective surface **4417** and are reflected at the third transmissive-reflective surface **4418**.

[0166] The fifth collimated optical signal and the sixth collimated optical signal exited from the fifth collimating lens **4425** and the sixth collimating lens **4426** each are reflected when incident on the second transmissive-reflective surface **4417** and are respectively multiplexed with the first sub composite optical signal and the third sub composite optical signal into the second sub composite optical signal and the fourth sub composite optical signal. The second sub composite optical signal and the fourth sub composite optical signal are reflected at the third transmissive-reflective surface **4418**.

[0167] The seventh collimated optical signal and the eighth collimated optical signal exited from the seventh collimating lens **4427** and the eighth collimating lens **4428** respectively transmit through the third transmissive-reflective surface **4418** and are respectively multiplexed with the second sub composite optical signal and the fourth sub composite optical signal into the first composite optical signal and the second composite optical signal. The first composite optical signal

and the second composite optical signal are respectively incident on the first polarizer **4408** and the second polarizer **4409** after being exited from the third transmissive-reflective surface **4418**.

[0168] It can be understood that a manner of multiplexing the first composite optical signal and the second composite optical signal into the merge composite optical signal and coupling the merge composite optical signal to the first fiber optic adapter **600**, so as to implement the light-emitting, is the same as the implementation manner (as shown in FIGS. **10A** and **10B**) in some embodiments above, and details will not be repeated herein.

[0169] FIG. **12** is a structural diagram of a lens group, a wavelength division multiplexer, and a supporting platform in an optical module, in accordance with some embodiments. FIG. **13** is a structural diagram of a supporting platform in an optical module, in accordance with some embodiments. As shown in FIGS. **12** and **13**, the light emitting assembly **400** further includes a supporting platform **450**. The supporting platform **450** is disposed in the light emitting cavity **403** and connected to the light emitting housing **420**. For example, the supporting platform **450** is disposed (e.g., adhered) on the bottom plate of the light emitting housing **420**. The first wavelength division multiplexer **443**, the second wavelength division multiplexer **444**, the first lens **445**, the second lens **446**, and the third lens **447** are disposed on the supporting platform **450**, so as to implement a passive coupling of the two wavelength division multiplexers and the lens group (i.e., the first lens **445**, the second lens **446**, and the third lens **447**) and reduce a coupling difficulty of a light emitting path.

[0170] In some embodiments, as shown in FIGS. **12** and **13**, the supporting platform **450** includes a supporting substrate **4501**, a lens fixing member, and a plurality of sub platforms. The supporting substrate **4501** includes a first portion **45011** and a second portion **45012**. The first portion **45011** is closer to the plurality of collimating lenses **442** than the second portion **45012**. The plurality of sub platforms are disposed on the first portion **45011**, and the lens fixing member is disposed on the second portion **45012**. The first wavelength division multiplexer **443** and the second wavelength division multiplexer **444** are fixed on the plurality of sub platforms, and the lens group is fixed on the lens fixing member.

[0171] In some embodiments, as shown in FIGS. **12** and **13**, the plurality of sub platforms include a first sub platform **451** and a second sub platform **456**, and the supporting platform **450** further includes a first mounting groove **452**, a first baffle **454**, a second mounting groove **457**, and a second baffle **459**.

[0172] The first sub platform **451** and the second sub platform **456** are disposed on the first portion **45011** of the supporting substrate **4501**. The first baffle **454** is disposed between the first sub platform **451** and the second sub platform **456**, the second baffle **459** is disposed on a side of the second sub platform **456** away from the first sub platform **451**, and the first baffle **454** is disposed in parallel with the second baffle **459**.

[0173] The first mounting groove **452** is disposed on the first sub platform **451**, and the second wavelength division multiplexer **444** is connected to the first sub platform **451** through the first mounting groove **452**. For example, the first mounting groove **452** includes a first groove wall **4521** and a second groove wall **4522** opposite to each other, the light incident surface of the second wavelength division multiplexer **444** abuts against the first groove wall **4521**, and the light exiting surface of the second wavelength division multiplexer **444** abuts against the second groove wall **4522**.

[0174] Similarly, the second mounting groove **457** is disposed on the second sub platform **456**, and the first wavelength division multiplexer **443** is connected to the second sub platform **456** through the second mounting groove **457**. For example, the second mounting groove **457** includes a third groove wall **4571** and a fourth groove wall **4572** opposite to each other. The light incident surface of the first wavelength division multiplexer **443** abuts against the third groove wall **4571**, and the light exiting surface of the first wavelength division multiplexer **443** abuts against the fourth groove wall **4572**.

[0175] In some embodiments, as shown in FIG. 13, the supporting platform 450 further includes a first glue accommodating groove 453. The first glue accommodating groove 453 is disposed at a bottom of the first mounting groove 452. The second wavelength division multiplexer 444 may be sealed with the bottom of the first mounting groove 452 by injecting glue into the first glue accommodating groove 453 and the first mounting groove 452, which is conducive to improving a sealing effect between the second wavelength division multiplexer 444 and the first sub platform 451, thereby improving an installation stability and reliability of the second wavelength division multiplexer 444.

[0176] For example, the supporting platform 450 further includes an annular protruding, and the annular protruding is disposed in the first mounting groove 452. The annular protruding extends from the bottom of the first mounting groove 452 to a direction away from the bottom of the first mounting groove 452 and defines the first glue accommodating groove 453 with the bottom of the first mounting groove 452.

[0177] In some embodiments, a side of the first mounting groove 452 facing toward the first baffle 454 is open, so that the glue may flow to the first mounting groove 452 from a surface of the first baffle 454 proximate to the first mounting groove 452, which is conducive to improving the reliability of the glue dispensing between the second wavelength division multiplexer 444 and the first mounting groove 452.

[0178] For example, a surface of the first baffle 454 proximate to the first mounting groove 452 is configured as a first slide 455. The top of the first slide 455 is closer to the second baffle 459 than the bottom of the first slide 455. In this way, when the glue is replenished to the first mounting groove 452 through the first slide 455, the glue may flow along the first slide 455 to the first mounting groove 452, so as to prevent the glue from accumulating at the bottom of the first baffle 454.

[0179] In some embodiments, as shown in FIG. 13, the supporting platform 450 further includes a second glue accommodating groove 458. The second glue accommodating groove 458 is disposed at a bottom of the second mounting groove 457. The first wavelength division multiplexer 443 may be sealed with the bottom of the second mounting groove 457 by injecting glue into the second glue accommodating groove 458 and the second mounting groove 457, which is conducive to improving a sealing effect between the first wavelength division multiplexer 443 and the second sub platform 456, thereby improving an installation stability and reliability of the first wavelength division multiplexer 443.

[0180] It will be noted that, the second glue accommodating groove 458 is formed in the same way as that of the first glue accommodating groove 453, and details will not be repeated herein.

[0181] A size of the first wavelength division multiplexer 443 is large, therefore, it is necessary to replenish the second glue accommodating groove 458 in order to avoid the reliability of the adhering between the first wavelength division multiplexer 443 and the second glue accommodating groove 458 being insufficient. A side of the second mounting groove 457 facing toward the second baffle 459 is open, so that the glue may flow into the second mounting groove 457 through the side surface of the second baffle 459.

[0182] In this way, the reliability of glue dispensing between the first wavelength division multiplexer 443 and the second mounting groove 457 may be improved.

[0183] For example, a surface of the second baffle 459 proximate to the second mounting groove 457 is configured as a second slide 4591. A top of the second slide 4591 is further away from the first baffle 454 than a bottom of the second slide 4591. In this way, when the glue is replenished to the second mounting groove 457 through the second slide 4591, the glue may flow along the second slide 4591 to the second mounting groove 457, so as to prevent the glue from accumulating at the bottom of the second baffle 459.

[0184] FIG. 14 is an enlarged view of cycle A in FIG. 13. As shown in FIGS. 13 and 14, the lens fixing member includes a lens mounting column, and the lens mounting column is configured to

mount and support the first lens **445** and the second lens **446**. For example, the first lens **445** and the second lens **446** abut against two side surfaces of the lens mounting column respectively and are fixedly connected with the supporting substrate **4501**.

[0185] In some embodiments, the lens mounting column is configured as a trigonal bulge **462**, and the supporting platform **450** further includes a trigonal groove **461**. The trigonal groove **461** is disposed on the second portion **45012** of the supporting substrate **4501**. The trigonal bulge **462** is disposed in the trigonal groove **461** and protrudes in a direction away from a bottom of the trigonal groove **461**.

[0186] The trigonal bulge **462** includes a first supporting surface **4621** and a second supporting surface **4622**. The first supporting surface **4621** faces toward the first wavelength division multiplexer **443**, and the exit surface **4451** of the first lens **445** is adhered to the first supporting surface **4621**, so that the first lens **445** receives the first composite optical signal exited from the first wavelength division multiplexer **443**. The second supporting surface **4622** faces toward the first fiber optic adapter **600**, the first incident surface **4461** of the second lens **446** is adhered to the second supporting surface **4622**, and the second lens **446** may receive the first composite optical signal passing through the first lens **445**.

[0187] It will be noted that, a thickness of the trigonal bulge **462** may be set according to the actual situation, as long as the light beams exited from the exit surface **4451** of the first lens **445** and the light beams entering the first incident surface **4461** of the second lens **446** are not shielded by the trigonal bulge **462**.

[0188] It can be understood that the trigonal groove **461** is disposed around the trigonal bulge **462**. In a first aspect, it is prone to form the trigonal bulge **462** on the supporting substrate **4501**. For example, it is convenient for tool withdrawal when turning the trigonal bulge (i.e., the trigonal groove **461** being a tool withdrawal groove). In a second aspect, due to a fact that a connection between the trigonal bulge **462** and the supporting substrate **4501** is fillet transition, so that the first lens **445** and the second lens **446** are not capable of fully contacting the trigonal bulge **462** when adhering to the trigonal bulge **462**. Therefore, by providing the trigonal groove **461**, the first lens **445** and the second lens **446** may fully contact the trigonal bulge **462**, which is conducive to improving installation accuracy of the first lens **445** and the second lens **446**. In a third aspect, when the first lens **445** and the second lens **446** are adhered to the trigonal bulge **462**, there may be excess glue overflow. The trigonal groove **461** may accommodate the excess glue, further improving the installation accuracy of the first lens **445** and the second lens **446**.

[0189] In some embodiments, as shown in FIGS. **12** and **13**, the supporting platform **450** further includes a linear groove **463**. The linear groove **463** is disposed on the second portion **45012** of the supporting substrate **4501**. An end of the linear groove **463** extends towards the first fiber optic adapter **600**, and another end of the linear groove **463** extends towards the light exiting surface of the second wavelength division multiplexer **444**. A bottom edge of the second incident surface **4471** of the third lens **447** contacts an edge of the linear groove **463** along a length direction, thereby fixing the third lens **447** at the linear groove **463**.

[0190] In some embodiments, the trigonal bulge **462** further includes a third supporting surface **4623**, and the third supporting surface **4623** faces toward the linear groove **463**, that is, the third supporting surface **4623** faces toward the second incident surface **4471**. In an orthogonal projection of the trigonal bulge **462** on the supporting substrate **4501**, sides where the first supporting surface **4621** and the third supporting surface **4623** are located are shorter sides of the orthogonal projection, and a side where the second supporting surface **4622** is located is a longer side of the orthogonal projection.

[0191] An angle between the first supporting surface **4621** and the second supporting surface **4622** satisfies a first preset condition and a second preset condition.

[0192] The first preset condition includes that when the first composite optical signal passing through the first lens **445** is incident on the first incident surface **4461** of the second lens **446**, the

incident angle of the first composite optical signal satisfies the preset angle that the angle selection film allows light beams to pass through.

[0193] The second preset condition includes that when the second composite optical signal reflected by the exit surface **4451** of the first lens **445** is incident on the first incident surface **4461** of the second lens **446**, the incident angle of the first composite optical signal satisfies the preset angle that the angle selection film allows the beam to pass through.

[0194] The inclination angle of the linear groove **463** with respect to the trigonal bulge **462** needs to be cooperated with the second supporting surface **4622** of the trigonal bulge **462**, so that the second composite optical signal reflected by the second incident surface **4471** of the third lens **447** is capable of being incident on the first incident surface **4461** of the second lens **446**, and the incident angle of the second composite optical signal is greater than the preset angle that the angle selection film allows light beams to pass through. In this way, the second composite optical signal may be reflected to the exit surface **4451** of the first lens **445** by the first incident surface **4461**, and then reflected to the first incident surface **4461** of the second lens **446** by the exit surface **4451** of the first lens **445**, and finally pass through the second lens **446**.

[0195] In some embodiments of the present disclosure, the first lens **445**, the second lens **446** and the third lens **447** implement the multiplexing of the first composite optical signal and the second composite optical signal by controlling the incident angles of the light beams. Therefore, the incident angles that the light beams are incident on the first lens **445**, the second lens **446**, and the third lens **447** are controlled by setting the relative angle between the trigonal bulge **462** and the linear groove **463**, so that the incident angle that the first composite optical signal is incident on the second lens **446** satisfies the preset angle that the first incident surface **4461** of the second lens **446** allows light beams to pass through, after the first composite optical signal passes through the first lens **445**, and the first composite optical signal may pass through the second lens **446**.

[0196] In addition, the second composite optical signal is reflected at the second incident surface **4471** of the third lens **447** and is reflected to the first incident surface **4461** of the second lens **446**. The second composite optical signal reflected to the first incident surface **4461** of the second lens **446** continues to reflect at the first incident surface **4461** and is reflected to the exit surface **4451** of the first lens **445**. The second composite optical signal is reflected at the exit surface **4451** of the first lens **445**, and the incident angle of the reflected second composite optical signal satisfies the preset angle that the first incident surface **4461** of the second lens **446** allows light beams to pass through, so that the reflected second composite optical signal may transmit through the second lens **446**.

[0197] In some embodiments of the present disclosure, the light emitting cavity **403** is provided with the supporting platform **450** therein. The supporting platform **450** is used to support and fix the first wavelength division multiplexer **443**, the second wavelength division multiplexer **444**, the first lens **445**, the second lens **446**, and the third lens **447**, and provide the optical path coupling space for these components.

[0198] The supporting platform **450** implements the passive coupling of the first wavelength division multiplexer **443** and the second wavelength division multiplexer **444** through the first sub platform **451** and the second sub platform **456**, so as to multiplex the eight collimated optical signals into two composite optical signals. In addition, the supporting platform **450** controls the incident angles that the light beams are incident on the first lens **445**, the second lens **446**, and the third lens **447** through the trigonal bulge **462** and the linear groove **463**, and multiplexes the first composite optical signal and the second composite optical signal into the merge composite optical signal, so that the passive coupling between the first lens **445**, the second lens **446**, and the third lens **447** is implemented, thereby effectively reducing the optical path coupling difficulty of the light emitting component **440**.

[0199] FIG. **15** is a perspective view of a laser, a heat dissipation component, and a ceramic adapter block in an optical module, in accordance with some embodiments. FIG. **16** is an exploded view of

a laser and a heat dissipation component in an optical module, in accordance with some embodiments. In some embodiments, as shown in FIGS. 15 and 16, the light emitting assembly 400 further includes a heat dissipation component 470. The heat dissipation component 470 is disposed in the light emitting cavity 403 and is configured to carry the lasers 441 and assist the lasers 441 in heat dissipation. The heat dissipation component 470 includes a plurality of chip on carrier substrates (COC Substrate) 472 and a plurality of heat sinks 473.

[0200] The plurality of chip on carrier substrates 472 are disposed at intervals on the bottom plate of the light emitting housing 420 and are configured to support the plurality of heat sinks 473 and transfer heat in the plurality of heat sinks 473 to the light emitting housing 420. For example, one chip on carrier substrate 472 may carry one or more heat sinks 473. The heat sink 473 is configured to support the laser 441 and dissipate the heat of the laser 441. For example, one or more heat sinks 473 may be adhered to one chip on carrier substrate 472 by means of glue.

[0201] In some embodiments, the heat dissipation component 470 further includes a thermoelectric cooler 471. The thermoelectric cooler 471 is disposed in the light emitting cavity 403 and is adhered to the bottom plate of the light emitting housing 420. For example, a bottom surface of the thermoelectric cooler 471 is adhered to the bottom plate of the light emitting housing 420 by means of glue. The thermoelectric cooler 471 is configured to support the plurality of chip on carrier substrates 472 and transfer heat in the plurality of chip on carrier substrate 472 to the light emitting housing 420.

[0202] In this way, the heat generated by the laser 441 in the light emitting process may be transferred to the heat sink 473, the chip on carrier substrate 472, and the thermoelectric cooler 471, which may effectively improve the heat dissipation efficiency of the laser 441.

[0203] For example, the heat dissipation component 470 includes one thermoelectric cooler 471, two chip on carrier substrates 472, and eight heat sinks 473. The two chip on carrier substrates 472 are disposed at an interval on the thermoelectric cooler 471. A position of one of the two chip on carrier substrates 472 corresponds to four lasers 441, and a position of another chip on carrier substrate 472 corresponds to the other four lasers 441. Four of the eight heat sinks 473 are disposed on the one chip on carrier substrate 472, and each of the four heat sinks 473 carries and fixes one laser 441. The other four heat sinks 473 are disposed on the another chip on carrier substrate 472, and each of the other four heat sinks 473 carries and fixes one laser 441.

[0204] In some embodiments, the heat sink 473 is provided with a pad, and the laser 441 is disposed on the pad. The laser driver 401 is provided with a corresponding pad, and the pad of the heat sink 473 is connected with the pad of the laser driver 401 through a gold wire. In this way, the laser driver 401 may send signals to the laser 441 through the gold wire and the pads to drive the laser 441, so as to emit optical signals.

[0205] In some embodiments, as shown in FIGS. 8 and 15, the light emitting component 440 further includes a plurality of optical detectors 480. The plurality of optical detectors 480 are disposed on the laser driver 401. One optical detector 480 corresponds to one laser 441 and is disposed on an optical path of a back of the laser 441. That is, a front of the laser 441 faces toward a corresponding collimating lens 442, so as to emit a light beam (i.e., optical signals) to the collimating lens 442, and the back of the laser 441 faces toward the corresponding optical detector 480. The optical detector 480 is configured to receive the light beam emitted from the back of the laser 441 and detect an optical power of the laser 441.

[0206] It can be understood that, in the light emitting process of the laser 441, the optical power of the light beam emitted by the laser 441 needs to be detected, and the optical power of the light beam emitted from the front of the laser 441 is the same as that of the light beam emitted from the back of the laser 441. Therefore, the optical detector 480 may be disposed on the optical path of the back of the laser 441, so as to detect the optical power of the light beam emitted from the back of the laser 441 and obtain the optical power of the light beam emitted from the front of the laser 441 according to the optical power of the light beam emitted from the back of the laser 441.

[0207] In some embodiments, the light emitting component **440** includes two laser drivers **401**. The four lasers **441** corresponding to the first wavelength division multiplexer **443** are fixed on four heat sinks **473** respectively, and the pads on the four heat sinks **473** are connected with the pads on one of the two laser drivers **401** through gold wires. The other four lasers **441** corresponding to the second wavelength division multiplexer **444** are fixed on the other four heat sinks **473** respectively, and the pads on the other four heat sinks **473** are connected with the pads on another laser driver **401** through gold wires.

[0208] FIG. **17** is a structural diagram of an optical detector and a laser driver in an optical module, in accordance with some embodiments. FIG. **18** is an exploded view of an optical detector in an optical module, in accordance with some embodiments. As shown in FIGS. **17** and **18**, the light emitting component **440** includes eight optical detectors **480**. Four optical detectors **480** are disposed on one laser driver **401**, and the other four optical detectors **480** are disposed on the other laser driver **401**. For example, the optical detectors **480** are fixed to the laser drivers **401** through welding.

[0209] In some embodiments, as shown in FIG. **18**, the optical detector **480** includes a bottom surface **488**, a top surface **481**, and a first side surface **482**, a second side surface **483**, a third side surface **4801**, and a fourth side surface **489** connected with edges of the bottom surface **488** and the top surface **481**, respectively. The first side surface **482**, the second side surface **483**, the third side surface **4801**, and the fourth side surface **489** are connected in sequence. The second side surface **483** (i.e., side surface) faces toward the laser **441**, and the second side surface **483** is a light inlet surface of the optical detector **480**.

[0210] The optical detector **480** further includes an anode **484** and a cathode **485**, and the anode **484** and the cathode **485** are disposed on the top surface **481**. The laser driver **401** further includes a common anode electrode **4011** (seeing FIG. **17**), and the common anode electrode **4011** is disposed on a side of the optical detectors **480** away from the lasers **441**. The anodes **484** of four optical detectors **480** are connected with one common anode electrode **4011** of one laser driver **401** through gold wires, and the anodes **484** of the other four optical detectors **480** are connected with one common anode electrode **4011** of the other laser driver **401** through gold wires.

[0211] The common anode electrode **4011** may be connected with the first flexible circuit board **410** through gold wires, and the cathode **485** of the optical detector **480** may further be connected with the first flexible circuit board **410** through gold wires.

[0212] It can be understood that, by providing the optical detector **480** and the common anode electrode **4011** on the laser driver **401**, the optical module **200** does not need to be provided with a support plate to support the optical detector **480** and the common anode electrode **4011**, which is conducive to saving an internal space of the optical module **200** and implementing the miniaturization of the optical module **200**.

[0213] In some embodiments, the optical detector **480** is configured as an oblique optical detector, that is, a portion of the second side surface **483** of the optical detector **480** is configured as an inclined plane **486**. An end of the inclined plane **486** proximate to the top surface **481** is closer to the laser **441** than an end of the inclined plane **486** proximate to the bottom surface **488**. That is, a size of the bottom surface **488** of the optical detector **480** is smaller than that of the top surface **481** in a direction (i.e., the N direction in FIG. **18**) from the second side surface **483** to the fourth side surface **489**.

[0214] In some embodiments, the optical detector **480** further includes a photosensitive surface **487**. The photosensitive surface **487** is located on the inclined plane **486**, that is, the photosensitive surface **487** is disposed obliquely. The photosensitive surface **487** corresponds to the light emitting surface of the back of the laser **441** and is configured to receive the light beam emitted from the back of the laser **441**.

[0215] It can be understood that, when the light beam emitted from the back of the laser **441** is incident on the photosensitive surface **487**, the light beam does not enter the optical detector **480**

entirely through the photosensitive surface **487**, but a portion of the light beam is reflected at the photosensitive surface **487**. Therefore, if the photosensitive surface **487** is perpendicular to an incident optical path of the light beam, the portion of the light beam may return to the laser **441** along the incident optical path of the light beam when the portion of the light beam is reflected at the photosensitive surface **487**, and the light emitting performance of the laser **441** may be affected. [0216] Therefore, in the optical module **200** provided by some embodiments of the present disclosure, the photosensitive surface **487** is disposed obliquely (i.e., the photosensitive surface **487** being disposed obliquely with respect to the incident optical path of the light beam). In this way, when the portion of the light beam is reflected at the photosensitive surface **487**, the reflected portion of the light beam does not coincide with the optical path of the incident light beam, and the reflected portion of the light beam will not return to the laser **441**. Thus, the optical detector **480** may be prevented from affecting the light emitting performance of the laser **441**, which is conducive to improving the stability of the light-emitting of the laser **441**.

[0217] FIG. **19** is an exploded view of a light emitting housing and a ceramic adapter block in an optical module, in accordance with some embodiments. FIG. **20** is a perspective view of a ceramic adapter block in an optical module, in accordance with some embodiments. As shown in FIG. **19**, the light emitting housing **420** includes a first inserting hole **421** (i.e., an inserting hole). The first inserting hole **421** is located in the side wall of the light emitting housing **420** away from the first fiber optic adapter **600** and is communicated with the outside of the light emitting assembly **400** (e.g., communicated with the inside of the housing). The first inserting hole **421** penetrates the light emitting housing **420** along a width direction (i.e., the S direction in FIGS. **7** and **19**) of the light emitting housing **420**, that is, the width of the light emitting housing **420** is the same as a width of the first inserting hole **421** along the S direction.

[0218] As shown in FIGS. **19** and **20**, the light emitting assembly **400** further includes a ceramic adapter block **490**, and a shape of the ceramic adapter block **490** matches an inner contour of the first inserting hole **421**. A portion of the ceramic adapter block **490** is inserted into the light emitting cavity **403** through the first inserting hole **421**, and another portion of the ceramic adapter block **490** is located outside the light emitting cavity **403**. A bottom surface of the ceramic adapter block **490** contacts the bottom plate of the light emitting housing **420**, and the top surface of the ceramic adapter block **490** contacts a top surface of the first inserting hole **421**. The ceramic adapter block **490** is configured to support and fix the laser driver **401** and is electrically connected to the laser driver **401** and the first flexible circuit board **410** respectively.

[0219] For example, the portion of the ceramic adapter block **490** inserted into the light emitting cavity **403** is provided with pads. The common anode electrode **4011** of the laser driver **401** is connected with the pads of the ceramic adapter block **490** through gold wires, and the cathode **485** of the optical detector **480** is connected with the pads of the ceramic adapter block **490** through gold wires. The pads of the laser driver **401** may also be connected with the pads of the ceramic adapter block **490** through gold wires. The portion of the ceramic adapter block **490** located outside the light emitting cavity **403** is electrically connected to the first flexible circuit board **410**.

[0220] In this way, the electrical signal, the working signal and other signals transmitted by the first flexible circuit board **410** may be transferred to the laser driver **401**, the optical detector **480**, the laser **441**, and other elements respectively through the ceramic adapter block **490**, so as to implement the normal operation of the laser driver **401**, the optical detector **480**, the laser **441**, and the other elements.

[0221] In some embodiments, the bottom surface of the ceramic adapter block **490** is connected with the bottom plate of the light emitting housing **420** by means of welding, and the top surface of the ceramic adapter block **490** is connected with the top surface of the first inserting hole **421** by means of welding.

[0222] In this way, a hermetic sealing in the light emitting housing **420** may be realized. In addition, referring to FIG. **7**, in a case where the ceramic adapter block **490** is welded to the light

emitting housing **420**, the end of the ceramic adapter block **490** proximate to the first fiber optic adapter **600** is connected with an inner wall of the first inserting hole **421**, and two sides of the ceramic adapter block **490**, along the S direction, are flush with two outer side surfaces of the light emitting housing **420**.

[0223] In some embodiments, as shown in FIGS. **19** and **20**, the ceramic adapter block **490** includes an adapter block body **491** and a boss **492**. A shape of the adapter block body **491** is matched with the first inserting hole **421**. The boss **492** is connected to the adapter block body **491** and is located on a side of the adapter block body **491** away from the first fiber optic adapter **600**. The boss **492** is configured to connect with the first flexible circuit board **410** and receive various signals sent by the circuit board **300** through the first flexible circuit board **410**, and the ceramic adapter block **490** transfers the various signals to the laser driver **401**, the optical detector **480**, the laser **441** and the other elements.

[0224] In some embodiments, as shown in FIG. **19**, the ceramic adapter block **490** further includes a first groove **493**, a second groove **494** and a third groove **495**. The first groove **493** is closer to the first fiber optic adapter **600** than the second groove **494** and is recessed downward compared with a bottom of the second groove **494**. The second groove **494** is closer to the first fiber optic adapter **600** than the third groove **495** and is recessed downward compared with a bottom of the third groove **495**. That is, the first groove **493**, the second groove **494** and the third groove **495** are arranged in a stepped manner.

[0225] The ceramic adapter block **490** further includes a plurality of pads **496**, and the plurality of pads **496** are disposed in the first groove **493**, the second groove **494** and the third groove **495**. The laser driver **401** is disposed in the first groove **493**. The pads on the laser driver **401**, the cathode **485** of the optical detector **480**, and the common anode electrode **4011** are connected with the plurality of pads **496** in the first groove **493**, the second groove **494**, and the third groove **495**, respectively, through gold wires, so as to implement the transfer of the various signals.

[0226] In some embodiments, the light transmitting assembly **400** includes two first flexible circuit boards **410**, and the boss **492** is connected with the circuit board **300** through the two first flexible circuit boards **410**. For example, an upper surface of the boss **492** is connected with one of the two first flexible circuit boards **410**, and a lower surface of the boss **492** is connected with another first flexible circuit board **410**, so as to implement the transmission of the various signals.

[0227] FIG. **21** is a structural diagram of a light emitting housing, a light emitting cover plate and a connecting block in an optical module, in accordance with some embodiments. As shown in FIGS. **19** and **21**, the light emitting assembly **400** further includes a connecting block **404**. The connecting block **404** is located on the side of the light emitting housing **420** proximate to the first fiber optic adapter **600**. An end of the connecting block **404** is connected with the outer surface of the light emitting housing **420**, and another end of the connecting block **404** is connected with the first fiber optic adapter **600**. The light emitting housing **420** further includes a through hole **422** (referring to FIG. **19**). The through hole **422** penetrates a side wall of the light emitting housing **420** connected with the connecting block **404** and is located at a position corresponding to the connecting block **404**. The through hole **422** is communicated with the light emitting cavity **403**. The merge composite optical signal emitted by the light emitting component **440** in the light emitting cavity **403** enters the connecting block **404** through the through hole **422**, and then enters the first fiber optic adapter **600** through the connecting block **404**.

[0228] Referring to FIGS. **19** and **21**, an upper end of the light emitting housing **420** is open, and the light emitting cover plate **430** is disposed at the opening, so as to realize a sealing of the light emitting assembly **400**. The light emitting cover plate **430** is welded with the light emitting housing **420** through a soldering machine, and the ceramic adapter block **490** is welded with the first inserting hole **421** (i.e., the light emitting housing **420**) through the soldering machine, so that the hermetic sealing in the light emitting housing **420** may be realized.

[0229] Generally, a thickness of the connecting block **404** is larger than that of the light emitting

housing **420** and the light emitting cover plate **430** after welding, that is, a top surface **4042** of the connecting block **404** protrudes from the light emitting cover plate **430**. Therefore, when the light emitting cover plate **430** is welded, the connecting block **404** may interfere with the soldering machine and affect the sealing between the light emitting cover plate **430** and the light emitting housing **420**.

[0230] In order to solve the above problem, in some embodiments, as shown in FIG. **21**, the connecting block **404** includes a connecting block body **4040** and an avoidance groove **4041**. The avoidance groove **4041** is proximate to the light emitting housing **420** and is recessed downward from the top surface **4042** of the connecting block **404**. A bottom of the avoidance groove **4041** is lower than the top of the light emitting housing **420** in a height direction (i.e., an L direction in FIG. **21**) of the light emitting housing **420**. Alternatively, the bottom of the avoidance groove **4041** is flush with the top surface of the light emitting housing **420**.

[0231] It will be noted that, a width of the avoidance groove **4041** may be set according to the actual situation, so as to prevent the connecting block **404** from interfering with the soldering machine when the light emitting housing **420** is welded with the light emitting cover plate **430**.

[0232] FIG. **22** is a partial sectional view of a light emitting assembly in an optical module, in accordance with some embodiments. As shown in FIG. **22**, the connecting block **404** further includes a photic hole **4043**. The photic hole **4043** penetrates the connecting block body **4040** along an arrangement direction from the first fiber optic adapter **600** to the light emitting housing **420**. The photic hole **4043** is communicated with the through hole **422**. In this way, the merge composite optical signal emitted by the light emitting component **440** in the light emitting cavity **403** may enter the first fiber optic adapter **600** through the through hole **422** and the photic hole **4043**, so as to realize the light-emitting.

[0233] In some embodiments, as shown in FIG. **22**, the converging lens **402** is installed in the connecting block **404**. For example, the connecting block **404** further includes a first accommodating groove **4044**, the first accommodating groove **4044** is located on the side of the connecting block **404** proximate to the first fiber optic adapter **600** and is communicated with the photic hole **4043**. The converging lens **402** is disposed in the first accommodating groove **4044**. In this way, the merge composite optical signal passing through the photic hole **4043** may be incident on the converging lens **402** and be coupled to an end surface of the optical fiber plug of the first fiber optic adapter **600** after being converged by the converging lens **402**.

[0234] In some embodiments, the flat lens **448** may be installed in the connecting block **404**. For example, the connecting block **404** further includes a second accommodating groove **4045**. The second accommodating groove **4045** is located on the side of the connecting block **404** connected with the light emitting housing **420** and is respectively communicated with the photic hole **4043** and the through hole **422**. The flat lens **448** is disposed in the second accommodating groove **4045**. In this way, the merge composite optical signal exited from the second lens **446** enters the flat lens **448** in the connecting block **404** through the through hole **422**. The flat lens **448** allows the merge composite optical signal to pass through and enter the photic hole **4043**.

[0235] In some embodiments, the flat lens **448** may also be installed in the side wall of the light emitting housing **420** connected with the connecting block **404**. In this case, the side wall is provided with the second accommodating groove **4045**, and the flat lens **448** is disposed in the second accommodating groove **4045**. In this way, the merge composite optical signal exited from the second lens **446** enters the flat lens **448** in the connecting block **404** through the through hole **422**. The flat lens **448** allows the merge composite optical signal to pass through and enter the photic hole **4043**.

[0236] In some embodiments, the connecting block **404** and the light emitting housing **420** are of an integral structure, or the connecting block **404** and the light emitting housing **420** are separate structures. In a case where the connecting block **404** and the light emitting housing **420** are separate structures, the side surface of the connecting block **404** may be adhered to the outer side of the light

emitting housing **420** by means of glue, so as to facilitate the installation of the flat lens **448** in the second accommodating groove **4045**.

[0237] In the optical module **200** provided by some embodiments of the present disclosure, the eight optical signals with different wavelengths are emitted by the eight lasers, and then converted to the eight collimated optical signals by the collimating lenses **442**. Afterwards, the eight collimated optical signals are multiplexed into the first composite optical signal and the second composite optical signal through the first wavelength division multiplexer **443** and the second wavelength division multiplexer **444** respectively, and then the first composite optical signal and the second composite optical signal are multiplexed into the merge composite optical signal through the first lens **445**, the second lens **446**, and the third lens **447**. The merge composite optical signal is coupled to the first fiber optic adapter **600**, thereby enabling simultaneous transmission of the plurality of optical signals with different wavelengths in one optical fiber.

[0238] In addition, in order to save the inner space of the optical module **200**, some embodiments of the present disclosure adopt the oblique optical detector **480**, so as to monitor the optical power of the laser **441**. The oblique optical detector **480** and the common anode electrode **4011** each are disposed on the laser driver **401**. The anode **484** of the optical detector **480** is connected with the common anode electrode **4011** through wiring, and the cathode **485** of the optical detector **480** is connected with the ceramic adapter block **490** through wiring. The laser driver **401** is connected with the ceramic adapter block **490** through wiring, and the ceramic adapter block **490** is connected with the first flexible circuit board **410**.

[0239] It can be understood that the various signals sent by the circuit board **300** may be transmitted to the laser driver **401**, the laser **441**, the optical detector **480**, and other photoelectric elements through the first flexible circuit board **410** and the ceramic adapter block **490**.

[0240] In some embodiments of the present disclosure, a parallel sealing process is used for the welding between the ceramic adapter block **490** and the light emitting housing **420**, and the welding between the light emitting cover plate **430** and the light emitting housing **420**. The avoidance groove **4041** is disposed on the connecting block **404** connecting the light emitting housing **420** and the first fiber optic adapter **600**. Therefore, it is possible to connect the first fiber optic adapter **600** and the light emitting housing **420** without affecting the parallel sealing. In this way, the plurality of optical signals with different wavelengths simultaneously transmit in one optical fiber may be implemented, and the space occupied by the light emitting assembly **400** in the optical module **200** may be reduced, which is conducive to the miniaturization design of the optical module **200**.

[0241] A person skilled in the art will understand that the scope of disclosure in the present disclosure is not limited to specific embodiments discussed above, and may modify and substitute some elements of the embodiments without departing from the spirits of this application. The scope of the application is limited by the appended claims.

Claims

1. An optical module, comprising: a circuit board; a light emitting assembly electrically connected with the circuit board, and the light emitting assembly comprising: a light emitting housing; a light emitting cover plate covered on the light emitting housing and defining a light emitting cavity with the light emitting housing; a plurality of lasers disposed in the light emitting cavity, and configured to emit a plurality of optical signals with different wavelengths; a plurality of wavelength division multiplexers disposed in the light emitting cavity and corresponding to the plurality of lasers; the plurality of wavelength division multiplexers being configured to multiplex the plurality of optical signals into a plurality of composite optical signals; a lens group disposed in the light emitting cavity, and configured to multiplex the plurality of composite optical signals into a merge composite optical signal; a plurality of laser drivers disposed in the light emitting cavity and

configured to drive the plurality of lasers; and a plurality of optical detectors disposed on the plurality of laser drivers and located on light-exit paths of back of the plurality of lasers; the plurality of optical detectors being configured to monitor optical powers of a plurality of light beams emitted from the back of the plurality of lasers.

2. The optical module according to claim 1, wherein at least one of the plurality of optical detectors comprises: a side surface facing toward a laser corresponding the at least one optical detector, and a portion of the side surface being configured as an inclined plane; a top of the inclined plane is closer to the laser than a bottom of the inclined plane; and a photosensitive surface disposed on the inclined plane, and a light beam emitted from back of the laser entering the optical detector through the photosensitive surface.

3. The optical module according to claim 1, further comprising a flexible circuit board, and the light emitting assembly and the lighting receiving assembly being coupled with the circuit board through the flexible circuit board; the lighting emitting assembly further comprises a ceramic adapter block, an end of the ceramic adapter block being located in the light emitting cavity, and another end of the ceramic adapter block being coupled with the flexible circuit board; the plurality of laser drivers being disposed on the ceramic adapter block; the plurality of lasers, the plurality of laser drivers and the plurality of optical detectors being connected with the ceramic adapter block.

4. The optical module according to claim 3, wherein any one of the plurality of optical detectors further comprises: a top surface; an anode; and a cathode, and the anode and the cathode being disposed on the top surface; wherein at least one of the plurality of laser drivers comprises a common anode electrode; and wherein the anode is connected with the common anode electrode, the common anode electrode is connected with the ceramic adapter block, and the cathode is connected with the ceramic adapter block.

5. The optical module according to claim 4, further comprising a fiber optic adapter, the fiber optic adapter being connected with the light emitting housing and configured to receive and transmit the merge composite optical signal; and the ceramic adapter block comprises: an adapter block body; a first groove, the plurality of laser drivers being disposed in the first groove; a second groove, the first groove being closer to the fiber optic adapter than the second groove, and being recessed downward compared with a bottom of the second groove; a third groove, wherein the first groove, the second groove and the third groove are disposed on the adapter block body; the second groove being closer to the fiber optic adapter than the third groove, and being recessed downward compared with a bottom of the third groove; and a pad, wherein the first groove, the second groove and the third groove each are provided with the pad; the cathode, the common anode electrode and the plurality of laser drivers are electrically connected with the pad.

6. The optical module according to claim 5, wherein the ceramic adapter block further comprises a boss, and the boss being disposed at a side of the adapter block body facing toward the circuit board; wherein the light emitting assembly comprises two first flexible circuit boards; and wherein a top surface of the boss is electrically connected with one of the two first flexible circuit boards, and a bottom surface of the boss is electrically connected with another of the two first flexible circuit boards.

7. The optical module according to claim 1, wherein the plurality of wavelength division multiplexers comprise: a first wavelength division multiplexer configured to receive a part of the plurality of optical signals emitted by a part of the plurality of lasers, and multiplex the part of optical signals into a first composite optical signal; and a second wavelength division multiplexer configured to receive another part of the plurality of optical signals emitted by another part of the plurality of lasers, and multiplex the another part of optical signals into a second composite optical signal.

8. The optical module according to claim 7, wherein the lens group comprise: a first lens disposed on a light-exit path of the first wavelength division multiplexer; a second lens disposed on a light-exit path of the first lens; a third lens disposed on a light-exit path of the second wavelength

division multiplexer; and wherein the first composite optical signal exited from the first wavelength division multiplexer transmits through the first lens and the second lens in turn; the second composite optical signal exited from the second wavelength division multiplexer is reflected from the third lens to the second lens, then reflected from the second lens to the first lens, and reflected from the first lens to the second lens, and transmitted through the second lens.

9. The optical module according to claim 8, wherein the second lens comprises an angle selection film, and the angle selection film being disposed on an incident surface of the second lens; the first lens comprises an optical filter, and the optical filter being disposed on an exit surface of the first lens; the third lens comprises a reflective film, and the reflective film being disposed on an incident surface of the third lens; wherein the first composite optical signal transmits through the optical filter; the second composite optical signal is reflected from the reflective film to the angle selection film, then reflected from the angle selection film to the optical filter, and reflected from the optical filter to the angle selection film, and transmits through the angle selection film; and wherein the first composite optical signal entering the second lens after transmitting through the first lens and the second composite optical signal entering the second lens after being reflected by the first lens are multiplexed into the merge composite optical signal in the second lens.

10. The optical module according to claim 8, further comprising a fiber optic adapter, the fiber optic adapter being connected with the light emitting housing and configured to receive and transmit the merge composite optical signal; and wherein the light emitting assembly further comprises at least one of: a converging lens disposed between the fiber optic adapter and the second lens, and configured to converge and couple the merge composite optical signal exited from the second lens to the fiber optic adapter; or, a converging lens disposed between the fiber optic adapter and the second lens, and configured to converge and couple the merge composite optical signal exited from the second lens to the fiber optic adapter; and a flat lens disposed in a side wall of the light emitting housing and located between the second lens and the converging lens; wherein the merge composite optical signal exited from the second lens transmits through the flat lens.

11. The optical module according to claim 1, wherein the light emitting assembly further comprises a plurality of collimating lenses; the plurality of collimating lenses being disposed between the plurality of lasers and the plurality of wavelength division multiplexers, and being configured to collimate the plurality of optical signals emitted by the plurality of lasers; wherein an end of a light incident surface of any one of the plurality of wavelength division multiplexers is closer to the plurality of collimating lenses than another end of the light incident surface; and wherein the any one of the plurality of wavelength division multiplexers comprises an incident optical port; and the incident optical port is disposed on the light incident surface of the any one of the plurality of wavelength division multiplexers and is configured to receive optical signals collimated by a part of the plurality of collimating lenses.

12. The optical module according to claim 7, wherein the light emitting assembly further comprises a supporting platform; the supporting platform is disposed in the light emitting cavity, and comprises: a supporting substrate; a first sub platform disposed on the supporting substrate; a first glue accommodating groove disposed on the first sub platform; the second wavelength division multiplexer being connected with the first sub platform through the first glue accommodating groove; a second sub platform disposed on the supporting substrate; a second glue accommodating groove disposed on the second sub platform; the first wavelength division multiplexer being connected with the second sub platform through the second glue accommodating groove; and a lens fixing member disposed on the supporting substrate; the lens group being fixed on the supporting platform through the lens fixing member.

13. The optical module according to claim 12, wherein the supporting platform further comprises: a first mounting groove disposed on the first sub platform; the second wavelength division multiplexer being disposed in the first mounting groove; the first glue accommodating groove being disposed in the first mounting groove, and the first glue accommodating groove being

connected with a bottom surface of the second wavelength division multiplexer; and a second mounting groove disposed on the second sub platform; the first wavelength division multiplexer being disposed in the second mounting groove; the second glue accommodating groove being disposed in the second mounting groove, and the second glue accommodating groove being connected with a bottom surface of the first wavelength division multiplexer.

14. The optical module according to claim 13, wherein the supporting platform further comprises: a first baffle disposed on the supporting substrate and located between the first sub platform and the second sub platform; a side surface of the first baffle facing toward the first sub platform being configured as a first slide, and a side of the first mounting groove facing toward the first slide being open; and a second baffle disposed on the supporting substrate and located at a side of the second sub platform away from the first baffle; a side surface of the second baffle facing toward the second sub platform being configured as a second slide, and a side of the second mounting groove facing toward the second slide being open.

15. The optical module according to claim 12, wherein the lens fixing member comprises: a lens mounting column, the first lens being fixedly connected with a side surface of the lens mounting column, so as to make an incident surface of a first lens face of the first lens toward the second sub platform; the second lens being fixedly connected with another side surface of the lens mounting column, so as to make an exit surface of the second lens face toward a fiber optic adapter, the fiber optic adapter being connected with the light emitting housing and configured to receive and transmit the merge composite optical signal;; and a linear groove, a third lens being disposed at the linear groove; wherein the lens mounting column is matched with the linear groove, so as to make relative angles between the first lens, the second lens and the third lens satisfy following conditions; wherein the first composite optical signal exited from the first wavelength division multiplexer transmits through the first lens and the second lens in turn; and wherein the second composite optical signal exited from the second wavelength division multiplexer is reflected from the third lens to the second lens, then reflected from the second lens to the first lens, and reflected from the first lens to the second lens, and transmits through the second lens.

16. The optical module according to claim 1, wherein the light emitting housing comprises: a through hole disposed in a side wall of the light emitting housing away from the circuit board; and an inserting hole disposed in a side wall of the light emitting housing proximate to the circuit board; wherein the light emitting assembly further comprises: a ceramic adapter block connected with the light emitting housing through the inserting hole and being sealed with the light emitting housing; and a connecting block, an end of the connecting block being connected to the light emitting housing and corresponding to a position of the through hole; the connecting block comprising: a connecting block body; and an avoidance groove disposed on the connecting block body, so as to avoid when the light emitting housing is welded with the light emitting cover plate; wherein the optical module further comprising a fiber optic adapter, the fiber optic adapter being connected with the light emitting housing and configured to receive and transmit the merge composite optical signal; the fiber optic adapter is connected with another end of the connecting block and is communicated with the through hole through the connecting block.

17. An optical module, comprising: a circuit board; a light emitting assembly electrically connected with the circuit board, and the light emitting assembly comprising: a light emitting housing; a light emitting cover plate covered on the light emitting housing and defining a light emitting cavity with the light emitting housing; a plurality of lasers disposed in the light emitting cavity, and configured to emit a plurality of optical signals with different wavelengths; a plurality of wavelength division multiplexers disposed in the light emitting cavity and corresponding to the plurality of lasers; the plurality of wavelength division multiplexers being configured to multiplex the plurality of optical signals into a plurality of composite optical signals; a lens group disposed in the light emitting cavity, and configured to multiplex the plurality of composite optical signals into a merge composite optical signal; a fiber optic adapter connected with the light emitting housing and

configured to receive and transmit the merge composite optical signal; wherein the light emitting housing comprises: a through hole disposed in a side wall of the light emitting housing away from the circuit board; and an inserting hole disposed in a side wall of the light emitting housing proximate to the circuit board; and wherein the light emitting assembly further comprises: a ceramic adapter block inserted into the light emitting cavity through the inserting hole and being sealed with the light emitting housing; a connecting block, an end of the connecting block being connected to the light emitting housing and corresponding to a position of the through hole; an avoidance groove being disposed on the connecting block body, so as to avoid when the light emitting housing is welded with the light emitting cover plate; and the fiber optic adapter is connected with another end of the connecting block and is communicated with the through hole through the connecting block.

18. The optical module according to claim 17, wherein the connecting block further comprises a top surface, and the avoidance groove being disposed on the top surface of the connecting block; a bottom of the avoidance groove being lower than a top of the light emitting housing along a height direction of the light emitting housing.

19. The optical module according to claim 17, wherein the connecting block comprises a connecting block body, and the avoidance groove is disposed on the connecting block body; the connecting block further comprises: a photic hole, an end of the photic hole being communicated with the through hole, and another end of the photic hole being communicated with the fiber optic adapter; and a first accommodating groove disposed at an end of the connecting block body proximate to the fiber optic adapter and communicated with the photic hole; wherein the light emitting assembly further comprises a converging lens, and the converging lens being disposed in the first accommodating groove.

20. The optical module according to claim 19, wherein the connecting block further comprises a second accommodating groove, the second accommodating groove being communicated with the through hole and the photic hole; the light emitting assembly further comprises a flat lens, and the flat lens being disposed in the second accommodating groove; wherein the second accommodating groove satisfies one of: the second accommodating groove being disposed at an end of the connecting block body proximate to the light emitting housing; or the second accommodating groove being disposed in the side wall of the light emitting housing proximate to the connecting block body.
